Manipulate – Facet – Reduce

Analysis: What, Why and How

- What is shown?
  - **Data** abstraction
- Why is the user looking at it?
  - **Task** abstraction
- How is it shown?
  - **Technique**: visual encoding and interaction



- Translation to vis vocabulary avoids domain-specific terms
  - Translation process iterative and often complex
- What-why-how analysis framework helps to think systematically about design space

How?

Encode

Arrange

- Express
- Separate
- Order
- Align
- Use



Map

from categorical and ordered attributes

- Color
  - Hue
  - Saturation
  - Luminance
- Size, Angle, Curvature, ...
- Shape
- Motion

Direction, Rate, Frequency, ...

Manipulate

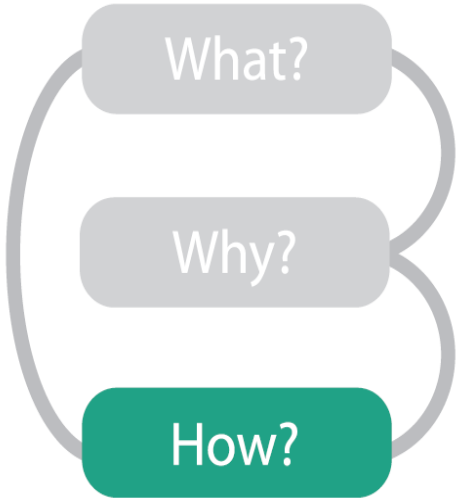
- Change
- Select
- Navigate

Facet

- Juxtapose
- Partition
- Superimpose

Reduce

- Filter
- Aggregate
- Embed



How?

Encode

Arrange

- Express
- Separate
- Order
- Align
- Use



Map

from categorical and ordered attributes

- Color
  - *Hue*
  - *Saturation*
  - *Luminance*
- Size, Angle, Curvature, ...
- Shape
- Motion

Direction, Rate, Frequency, ...



Manipulate

- Change
- Select
- Navigate

Facet

- Juxtapose
- Partition
- Superimpose



Reduce

- Filter
- Aggregate
- Embed



Manipulate

Change



Select



Navigate



Facet

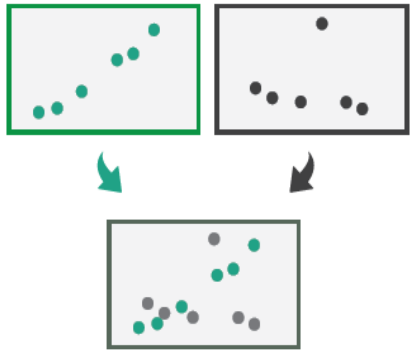
Juxtapose



Partition



Superimpose

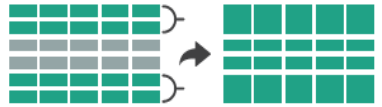


Reduce

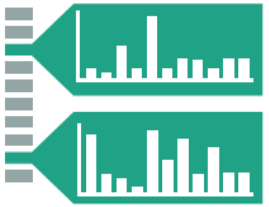
Filter



Aggregate



Embed



Manipulate

Change



Select



Navigate



Facet

Juxtapose

Partition

Superimpose

Reduce

Filter

Aggregate

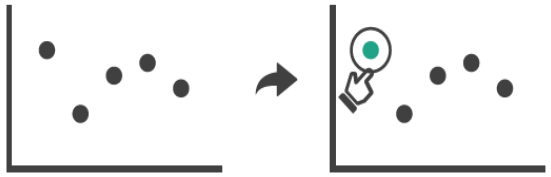
Embed

Manipulate

Change over Time



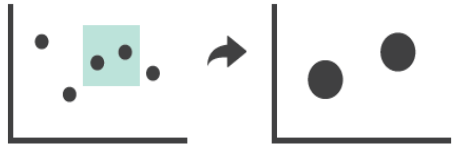
Select



Navigate

Item Reduction

- *Zoom*
  Geometric or *Semantic*



- *Pan/Translate*



- *Constrained*

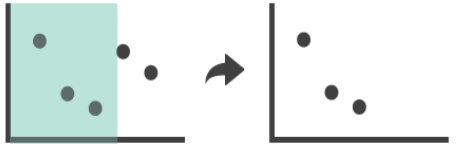


Attribute Reduction

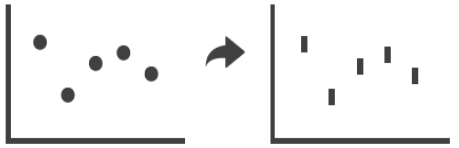
- *Slice*



- *Cut*



- *Project*



Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014. Chapter 11

Navigation Standard Solution: Scrolling (Detail-only window)

- Single window with horizontal and vertical panning to access a larger virtual screen
- Issues
  - Only single focus
  - Efficient scrolling requires cognitive map
    - No overview
    - Relationship between the current focus and the desired focus?
    - Must mentally map relationship between current and next foci to the slider positions on the scroll bar
  - Works well only if zoom factor is relatively small



Overview First, Zoom and Filter, Details on Demand

- Influential mantra from Ben Shneiderman

- What is needed
  - Overview (summary)
  - Focus view
  - Support for navigation

[The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. Shneiderman. Proc. IEEE Visual Languages, pp. 336–34 1996.]

[Search, Show Context, Expand on Demand: Supporting Large Graph Exploration with Degree-of-Interest. van Ham and Perer. IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis 2009) 15:6 (2009), 953–960.]

Overview + Detail

- Provide both overview and detail displays
- Two ways to combine:
  - Time - Alternate between overview and detail sequentially
  - Space - Use different portions of the screen (eye beats memory)



Overview and detail (from *Civilization II* game)

Plaisant et al Article

- Good introduction to overview/detail issue
- Application concern: viewing and editing large images
- Expanding the notion of the one-dimensional scroll bar: zooming, diagonal panning, multiple detailed views
- List of visualization/interaction solutions

Image Browsers: Taxonomy and Design Guidelines. C. Plaisant, D. Carr, B. Shneiderman *IEEE Software* 12(2) March 1995 pp 21-32

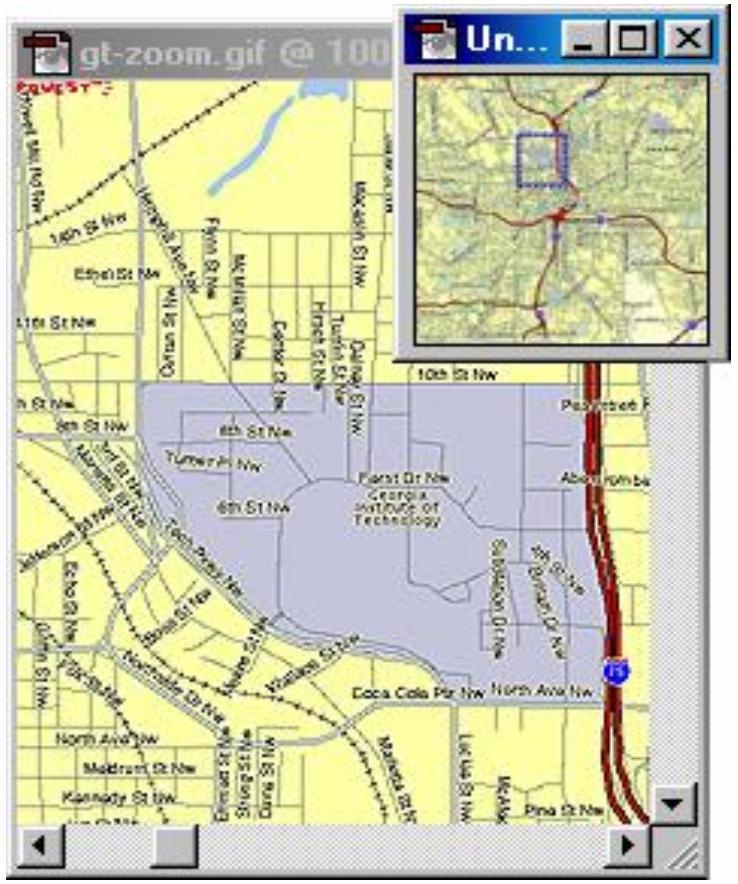
One Window With Zoom & Replace

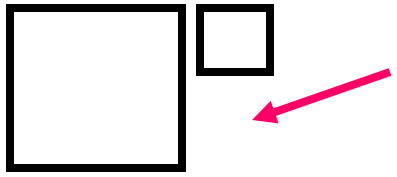
- Global view with selectable zoom area which then becomes entire view
- Variations can let users pan and adjust zoomed area and adjust levels of magnification
- Context switch can be disorienting
- Example: CAD/CAM



Single Coordinated Pair

- Combined display of the overview and local magnified view (separate views)
- Some implementations reserve large space for overview; others for detail
- Issue: How big are different views and where do they go?

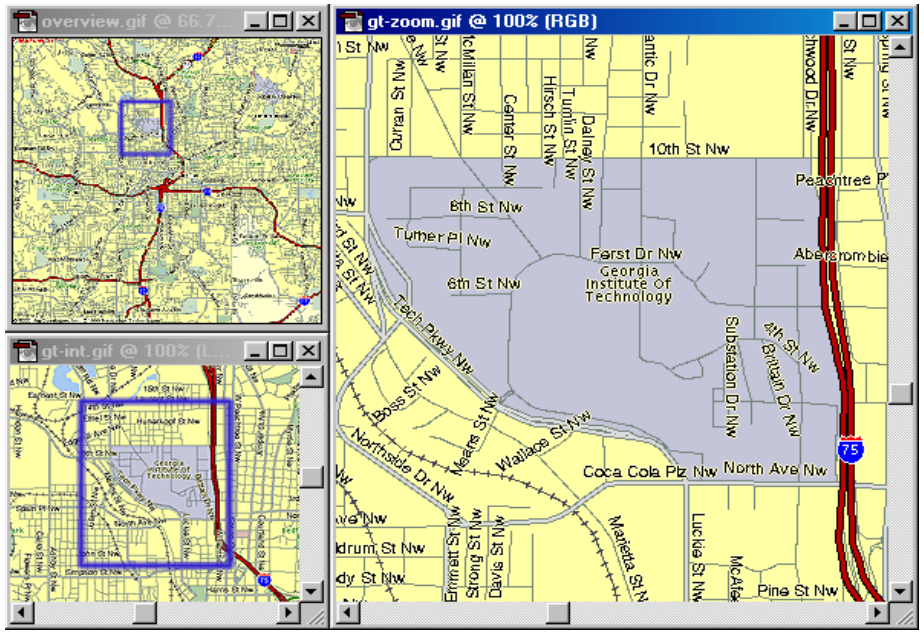




Wasted space

Tiled Multilevel Browser

- Combined global, intermediate, and detail views
- Views do not overlap
- Good implementations closely relate the views, allowing panning in one view to affect others



Free Zoom And Multiple Overlap

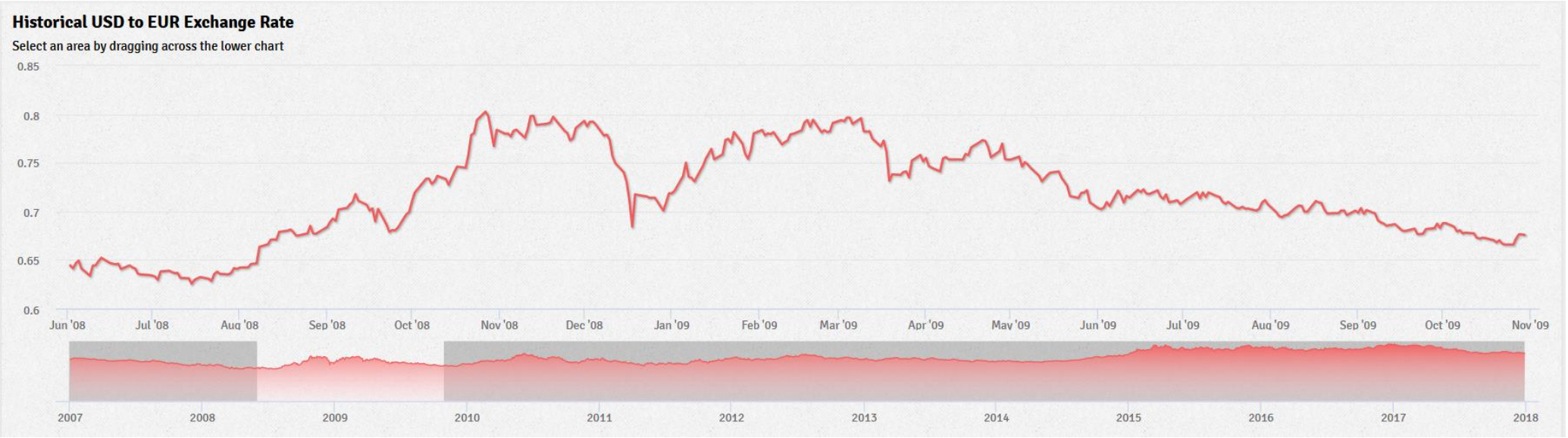
- Overview presented first; user selects area to zoom and area in which to create detailed view

- Flexible layout, but users must perform manual window management



Overview and Detail for Time Series Data

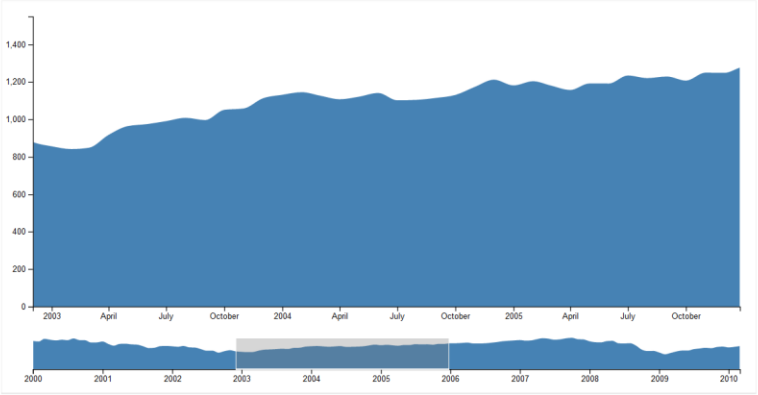
https://www.highcharts.com/demo/dynamic-master-detail



Mike Bostock's Block 34f08d5e11952a80609169b7917d4172
Updated April 17, 2019

Popular / About

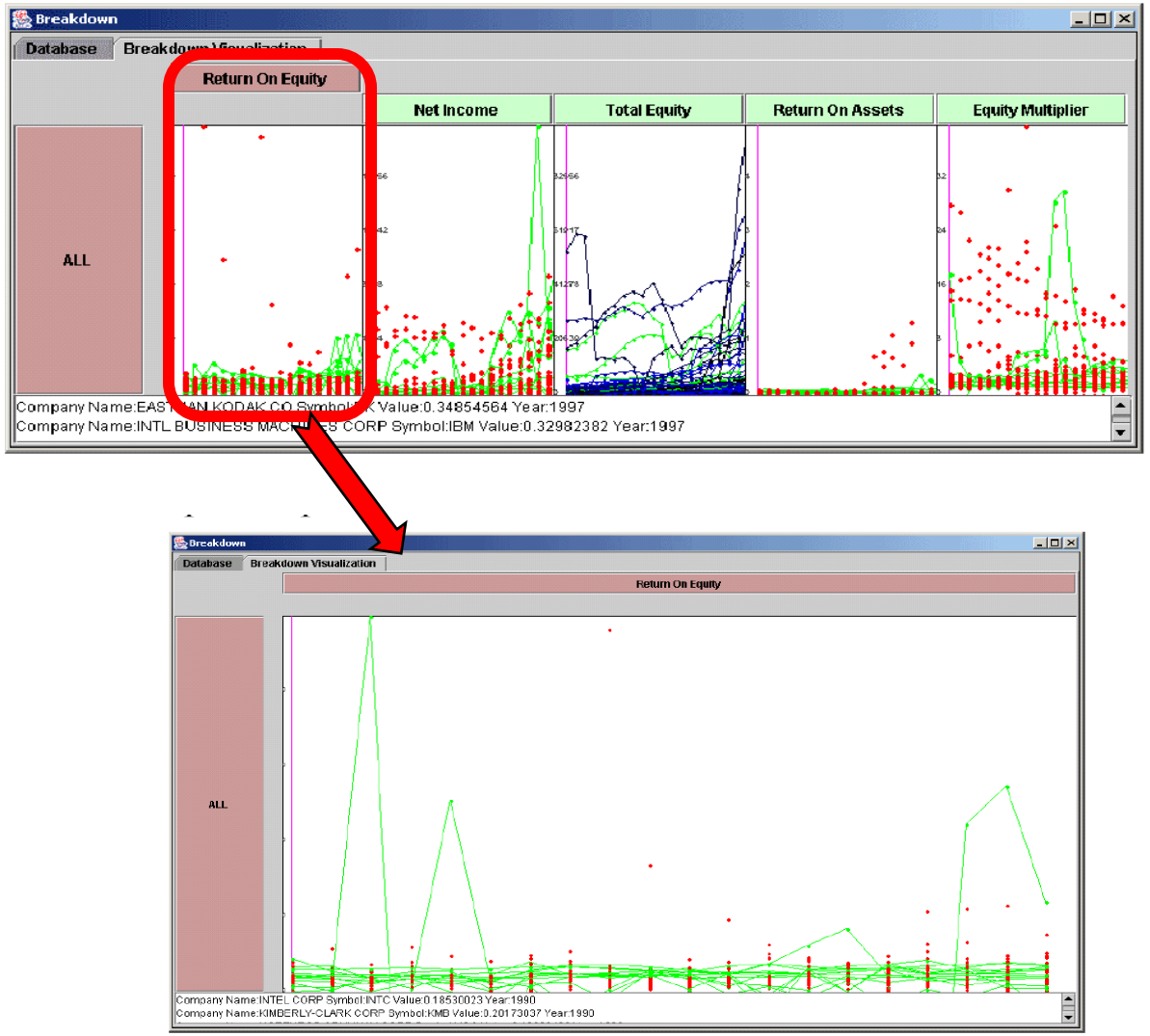
Brush & Zoom



https://bl.ocks.org/mbostock/34f08d5e11952a80609169b7917d4172

Overview and Detail: Drill-down

- Manually select individual item or smaller set of items from a display for a more detailed view/analysis



Conklin et al.
InfoVis2002

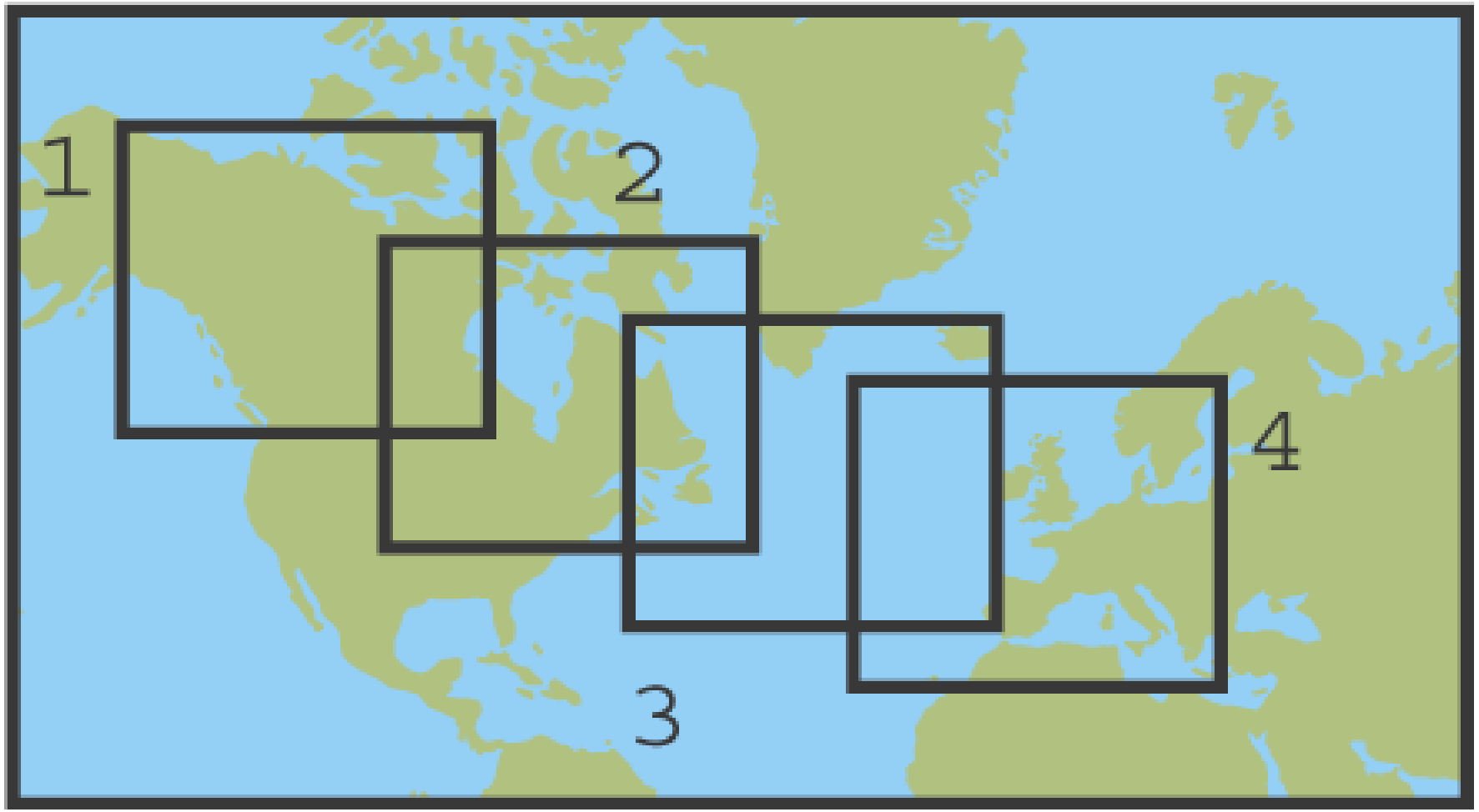
Panning and Zooming

- Panning
  - Smooth movement of camera across scene (or scene moves and camera stays still)
- Zooming
  - Increasing or decreasing the magnification of the objects in a scene

- Useful for changing focal point
- ZUI = Zoomable User Interface

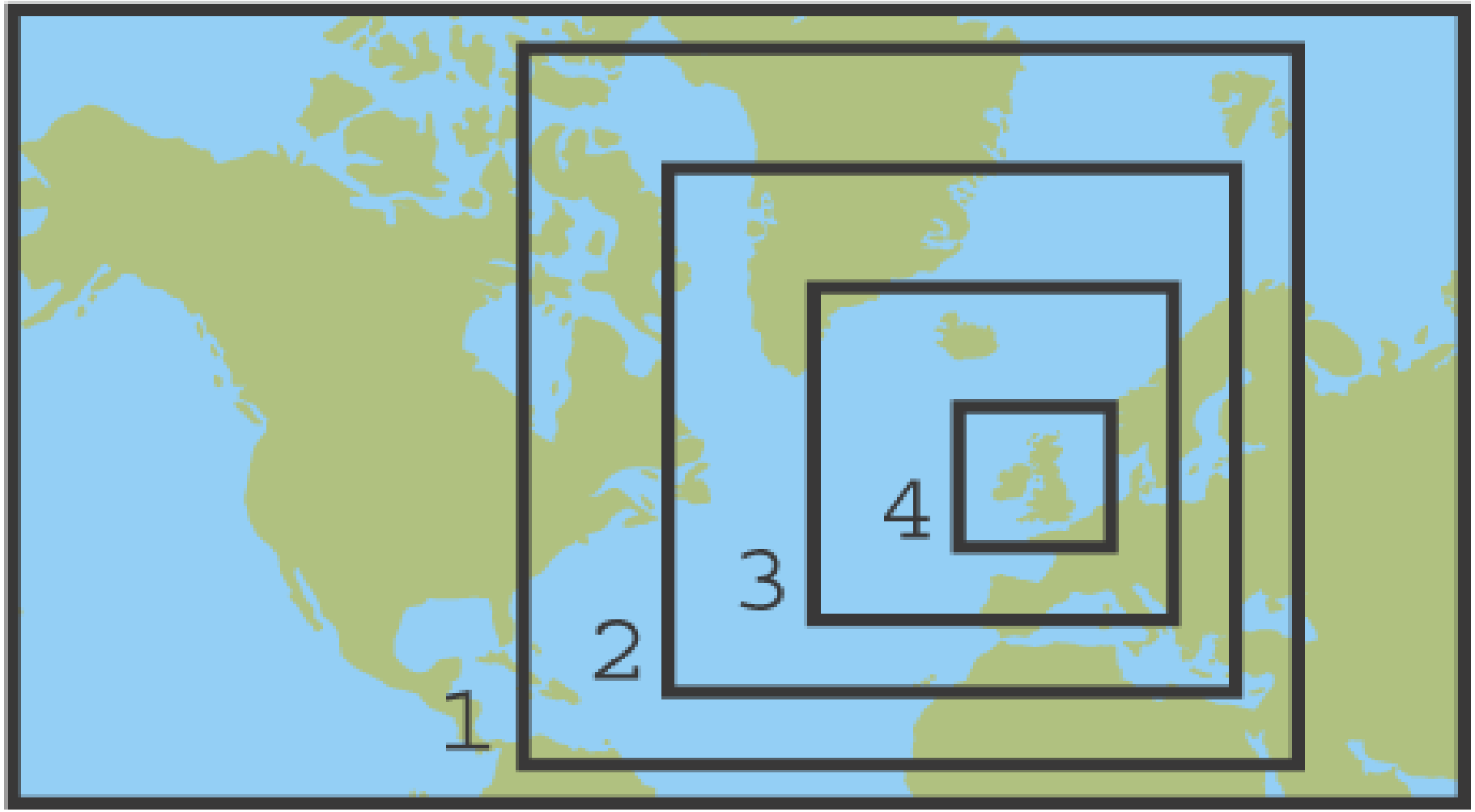
Panning

- Smooth movement of a viewing window over a 2D image



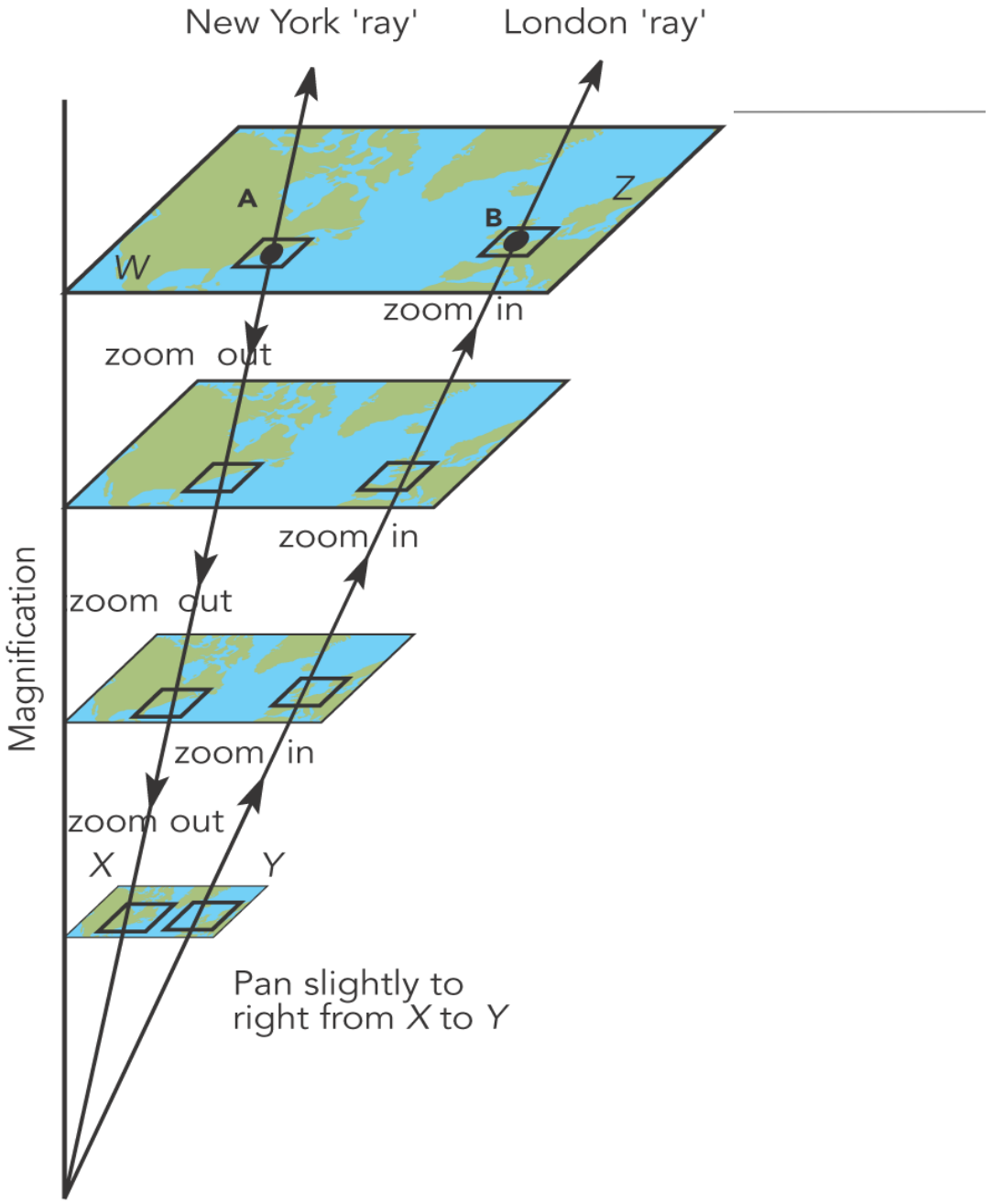
Zooming

- Increasing/ or decreasing magnification



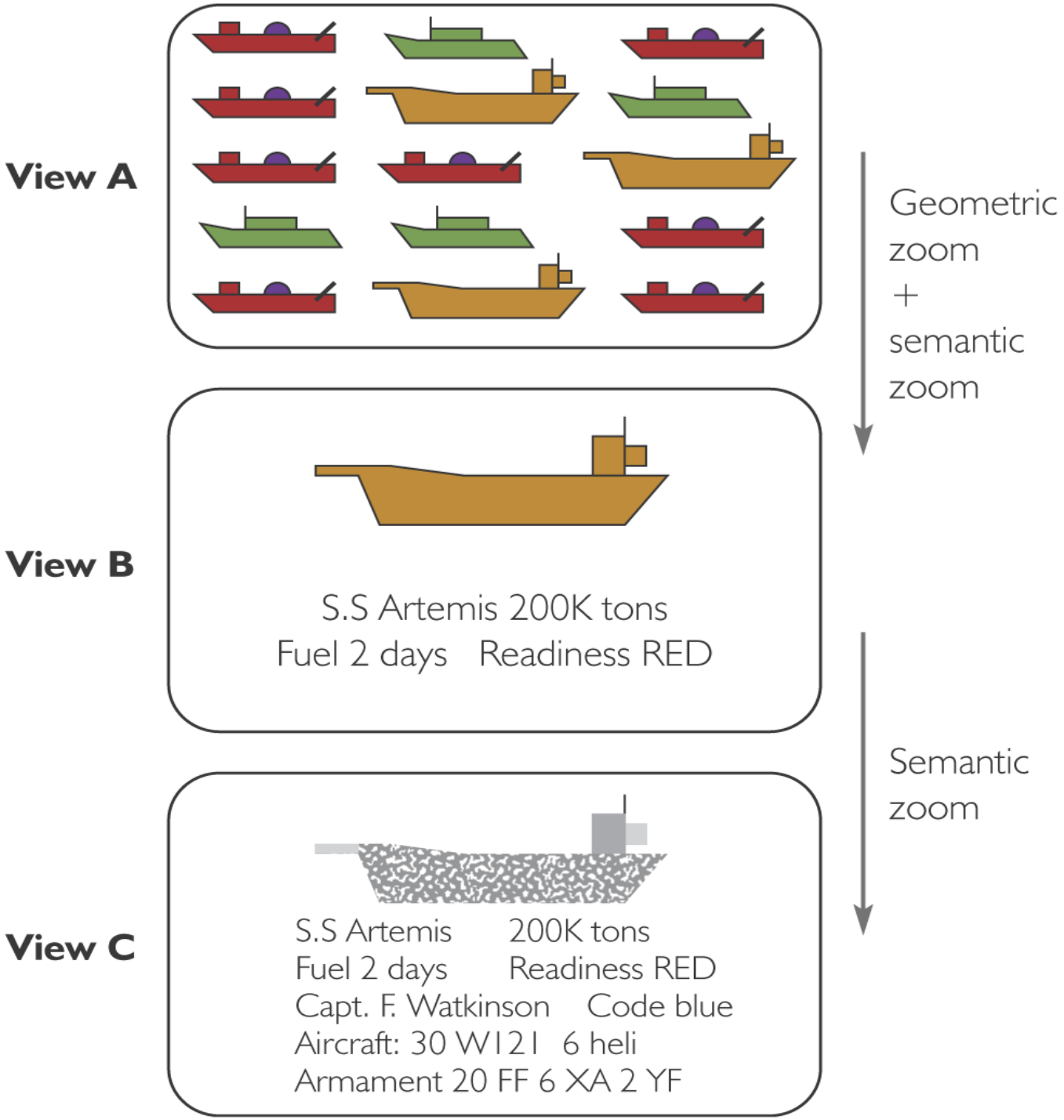
Combined Zooming and Panning

- Efficient task sequence
  - Zoom back
  - Pan
  - Zoom in



Geometric + Semantic Zoom

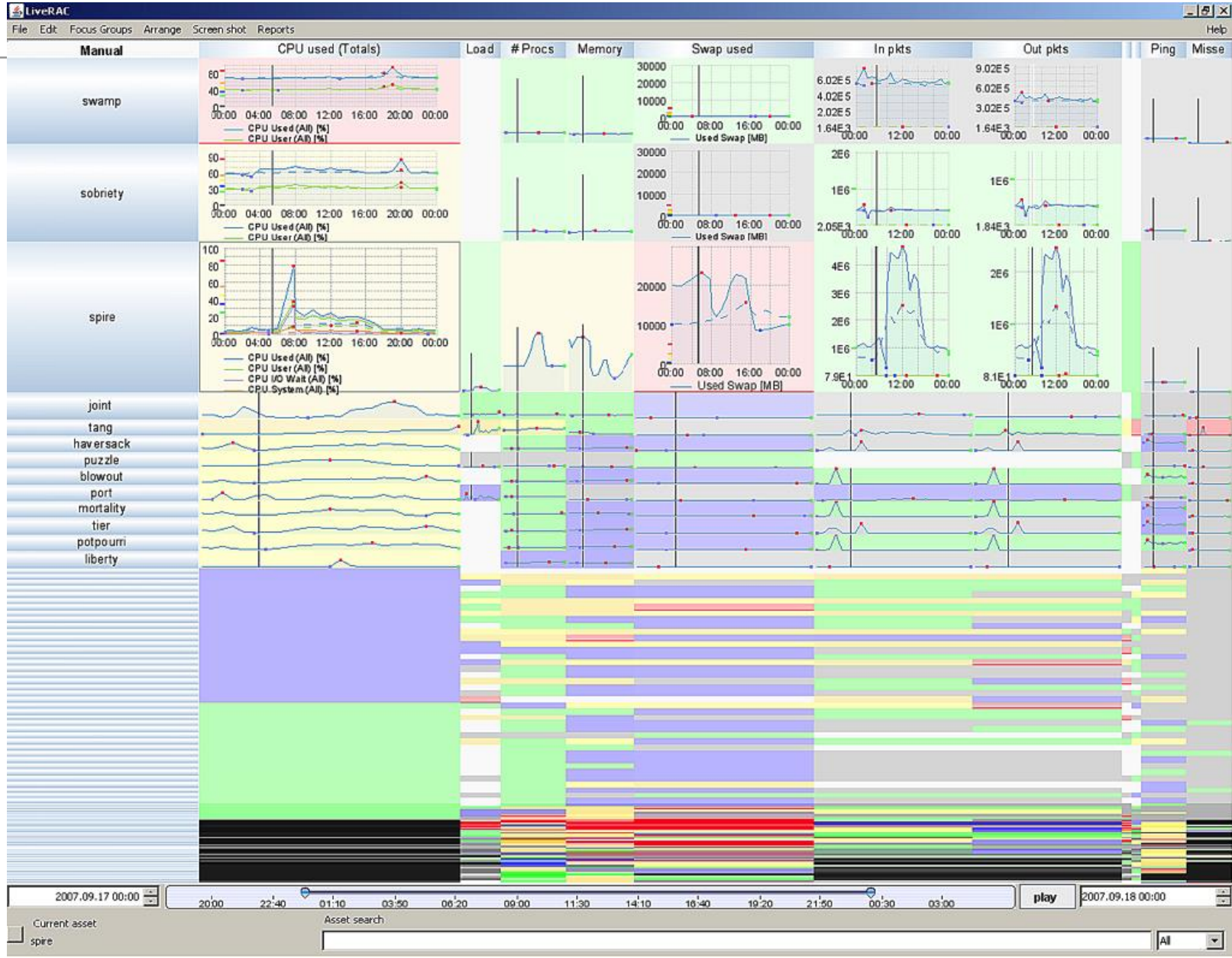
- Often just simple zooming is not all that useful
- Change representation based on zooming level!



Semantic Zooming

- Alternative to geometric zoom
- Resolution-aware layout adapts to available space
- Goal: Legible at multiple scales

- Example: visual encoding changes from bottom to top
  - Colored box
  - Simple line chart
  - Full chart: axes and tickmarks



System: LiveRAC

LiveRAC - Interactive Visual Exploration of System Management Time-Series Data. McLachlan, Munzner, Koutsofios, and North. Proc. ACM Conf. Human Factors in Computing Systems (CHI), pp. 1483–1492, 2008.

Manipulate

Facet

Reduce

Change

Juxtapose



Filter

Select

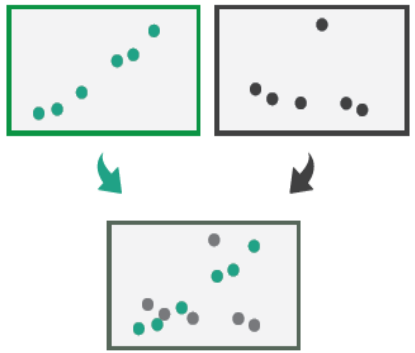
Partition



Aggregate

Navigate

Superimpose



Embed

Quiz

Drag the words to complete Shneiderman's influential mantra.

_____ first, _____ and filter, _____ on demand

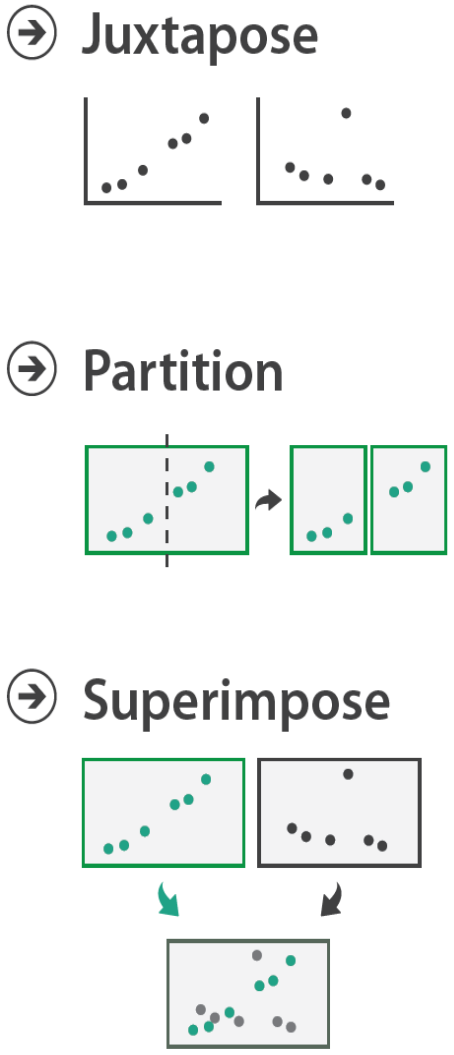
details | zoom | overview

Check

Facet (Split Up)

- Handle visual complexity
  - Juxtaposing coordinated views side by side
    - Show different facets or views of the data
  - Superimposing layers in a single view
    - Requires splitting up in layers
      e.g. layers in a map: streets vs. water, land areas, …

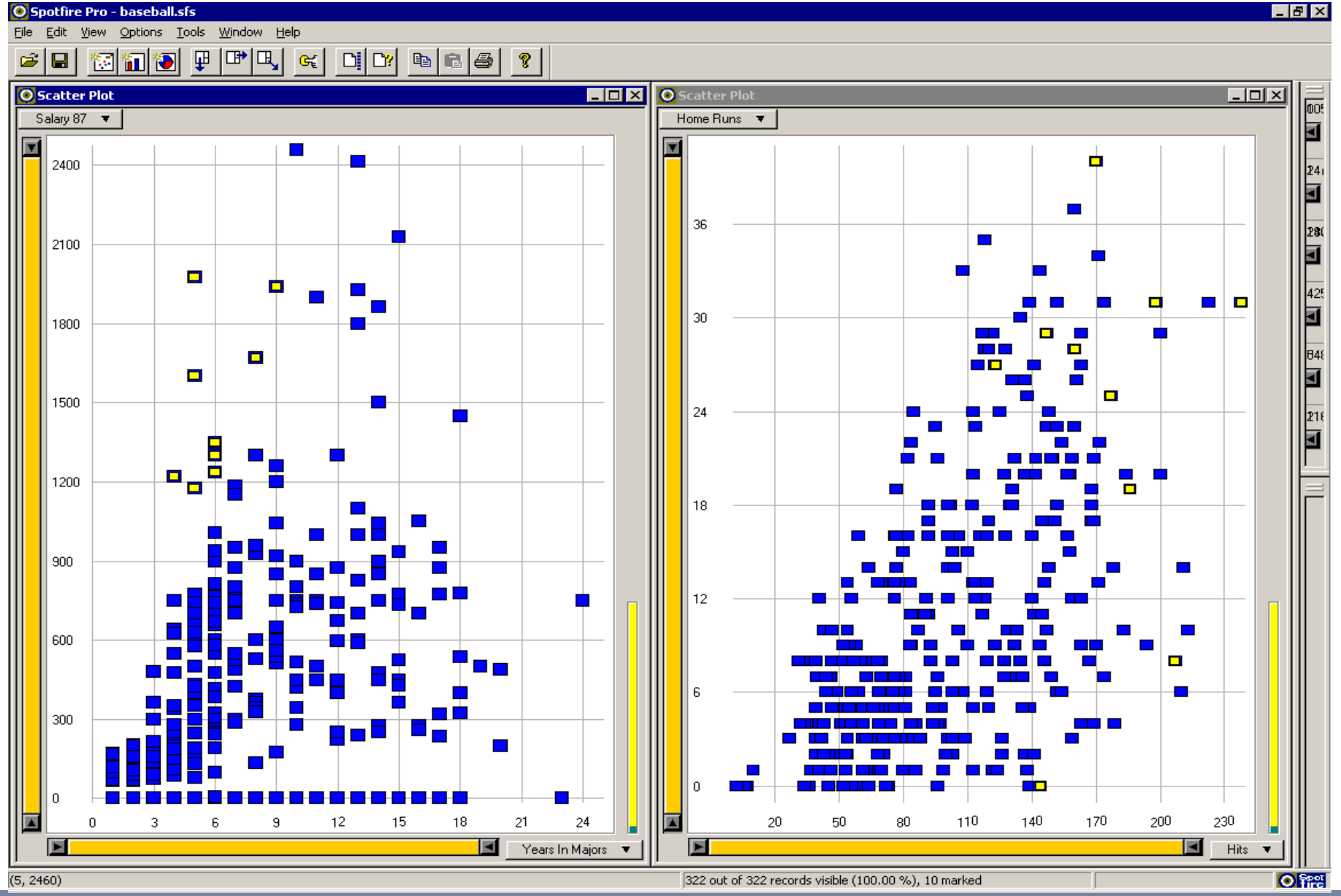




Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014. Chapter 12

Juxtapose: Coordination Through Brushing and Linking

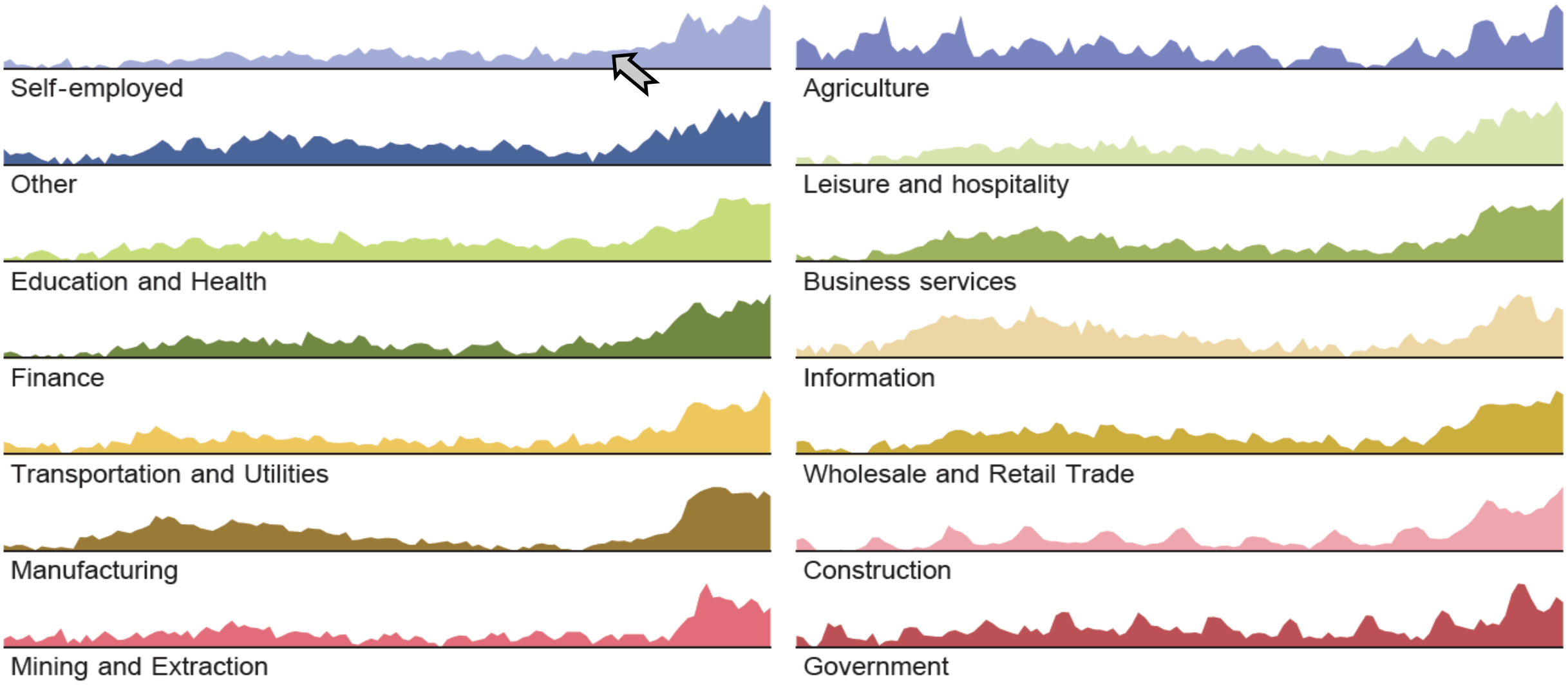
Select or designate/specify value, then see matching items elsewhere on the display



Example: 4D data in linked 2D + 2D display

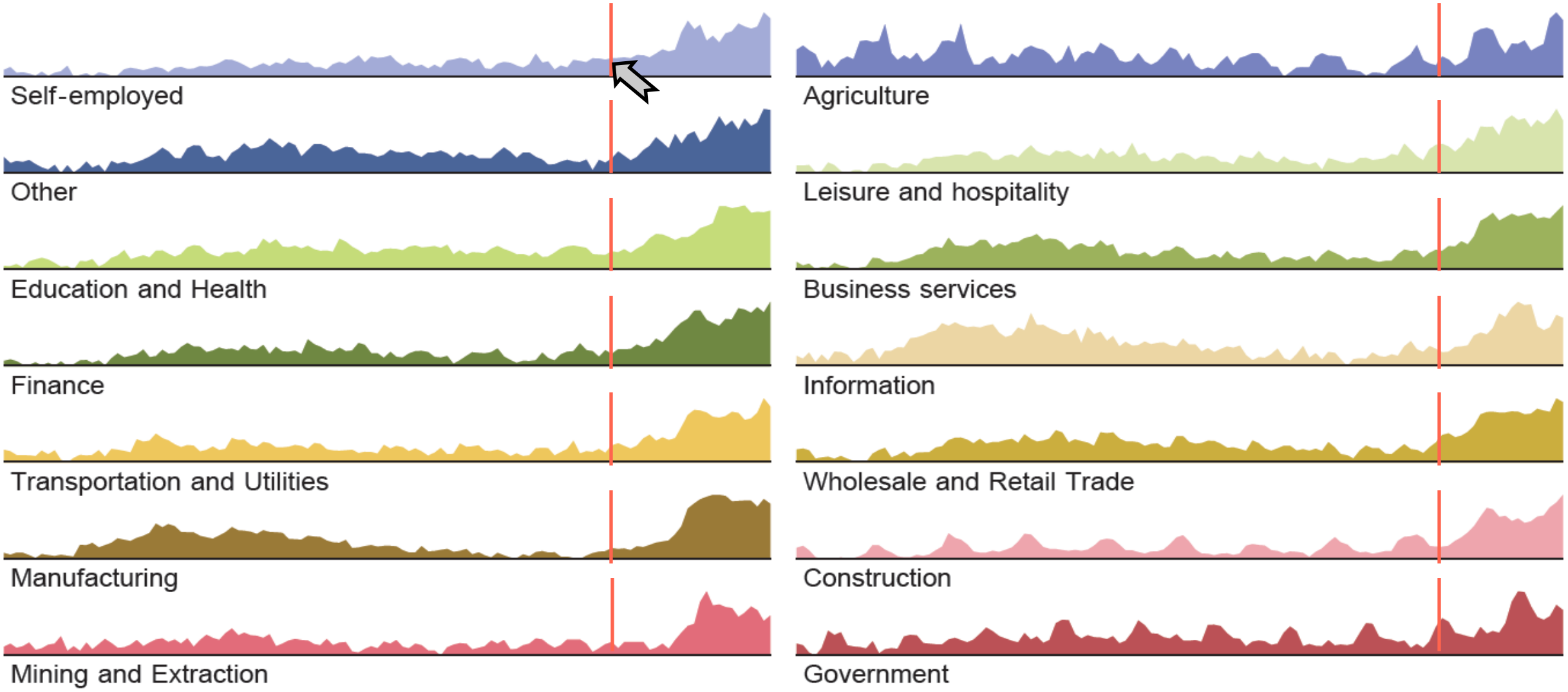
Items are major league baseball players.

Juxtapose: Small Multiples



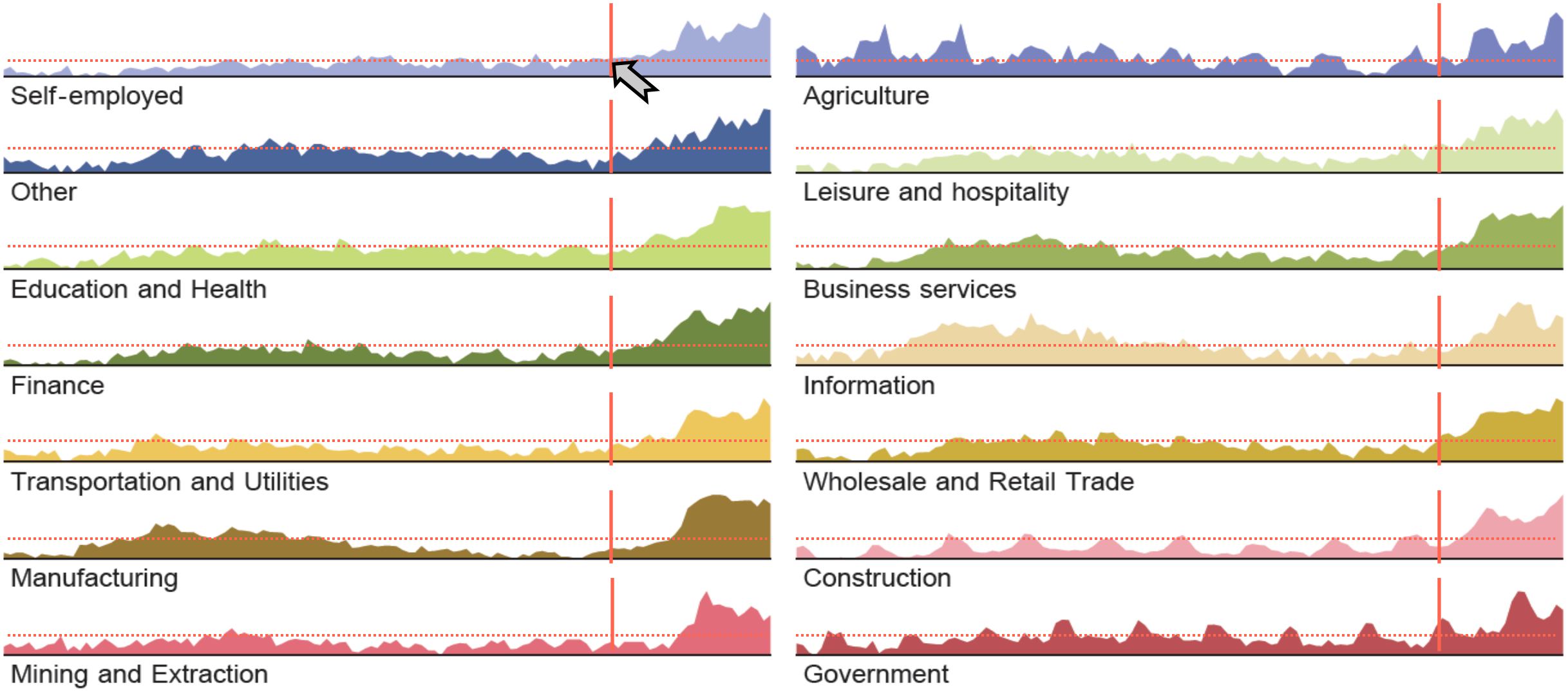
n e.g. through time slider Unemployed U.S. workers by industry, 2000–2010 Data source: U.S. Bureau of Labor Statistics http://hci.stanford.edu/jheer/files/zoo/ex/time/multiples.html

Juxtapose: Small Multiples



n e.g. through time slider Unemployed U.S. workers by industry, 2000–2010 Data source: U.S. Bureau of Labor Statistics http://hci.stanford.edu/jheer/files/zoo/ex/time/multiples.html

Juxtapose: Small Multiples



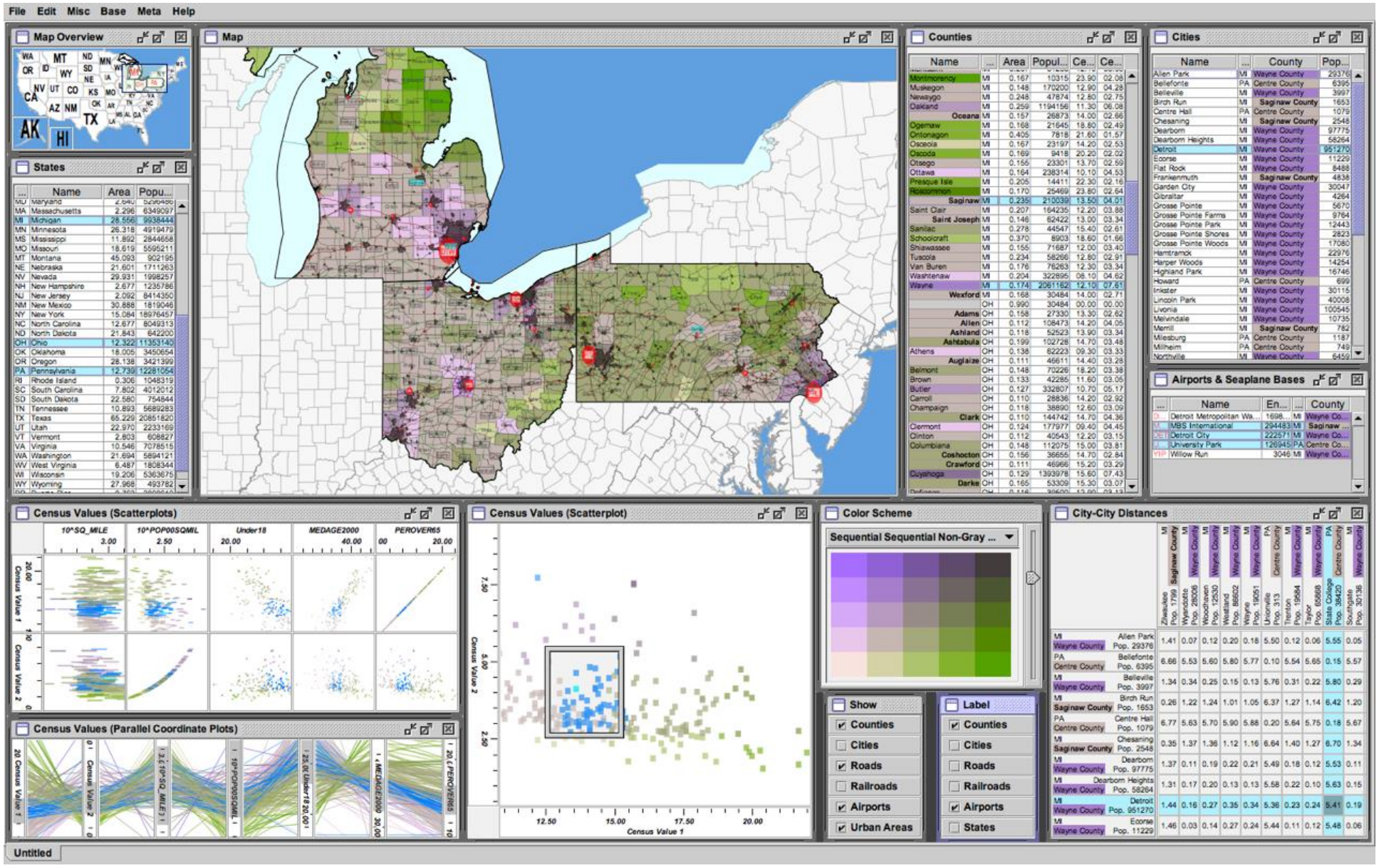
n e.g. through time slider Unemployed U.S. workers by industry, 2000–2010 Data source: U.S. Bureau of Labor Statistics http://hci.stanford.edu/jheer/files/zoo/ex/time/multiples.html

Small Multiples

- Can be constructed for plots, area graphs, bar charts, …
- Also interesting as linked views where coordinated interaction takes place across all individual visualizations
- Separate visualizations can be more effective than overlaying all data in a single plot

Coordinated Multiple-View Display Vis System: Improvise

- Investigate power of multiple views
  - The system is pushing the limits on view count and interaction complexity

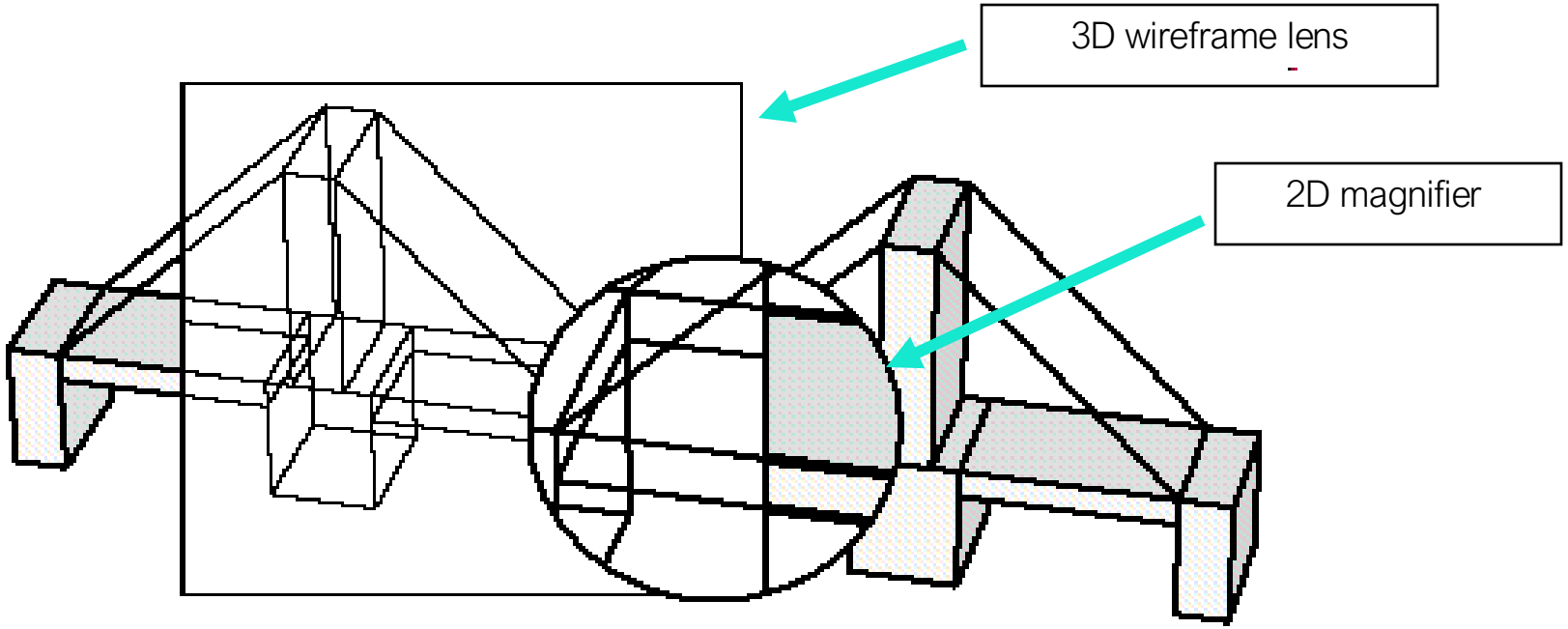


[Building Highly-Coordinated Visualizations In Improvise. Weaver. Proc. IEEE Symp. Information Visualization (InfoVis), pp. 159–166, 2004.]

Superimpose Local Lens: Lenses

Magic Lenses

- Regions/objects that can be moved around on display
- Semantically different content may be displayed in the lens e.g. semantic zoom!



Toolglass and Magic Lenses: The See-Through Interface, Siggraph 1993
Eric A. Bier, Maureen C. Stone, Ken Pier, William Buxton, Tony D. DeRose http://www2.parc.com/istl/projects/MagicLenses/93Siggraph.html

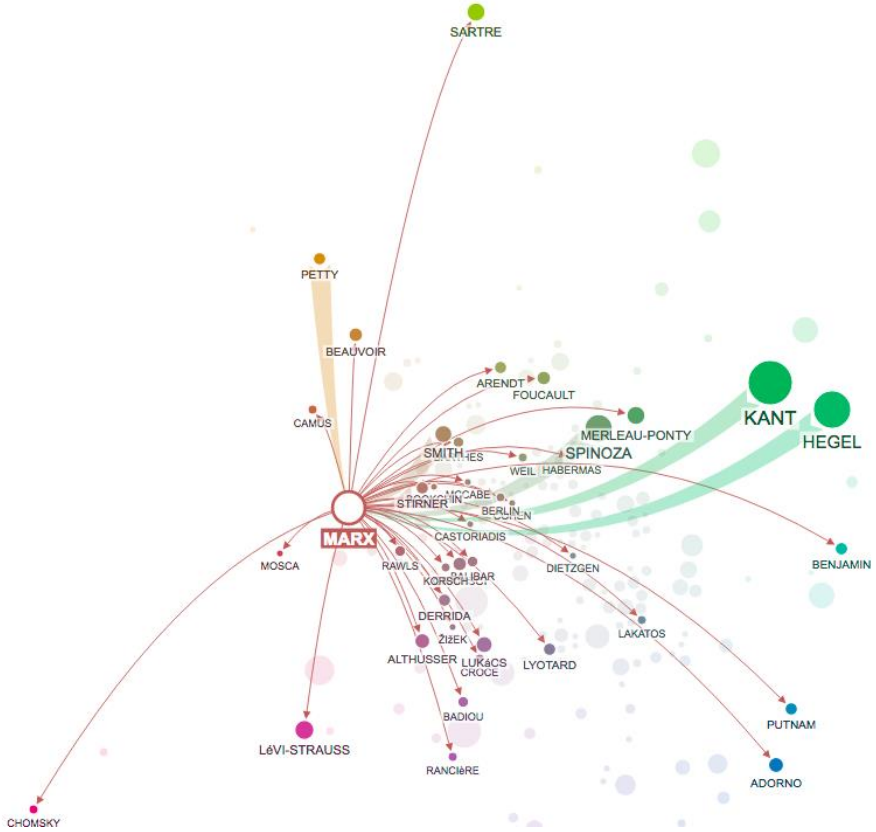
Superimpose Local Lens: Bifocal Magnified

- “Magnifying glass” zoomed image floats over overview image
- Adjacent objects are obscured by the zoomed window

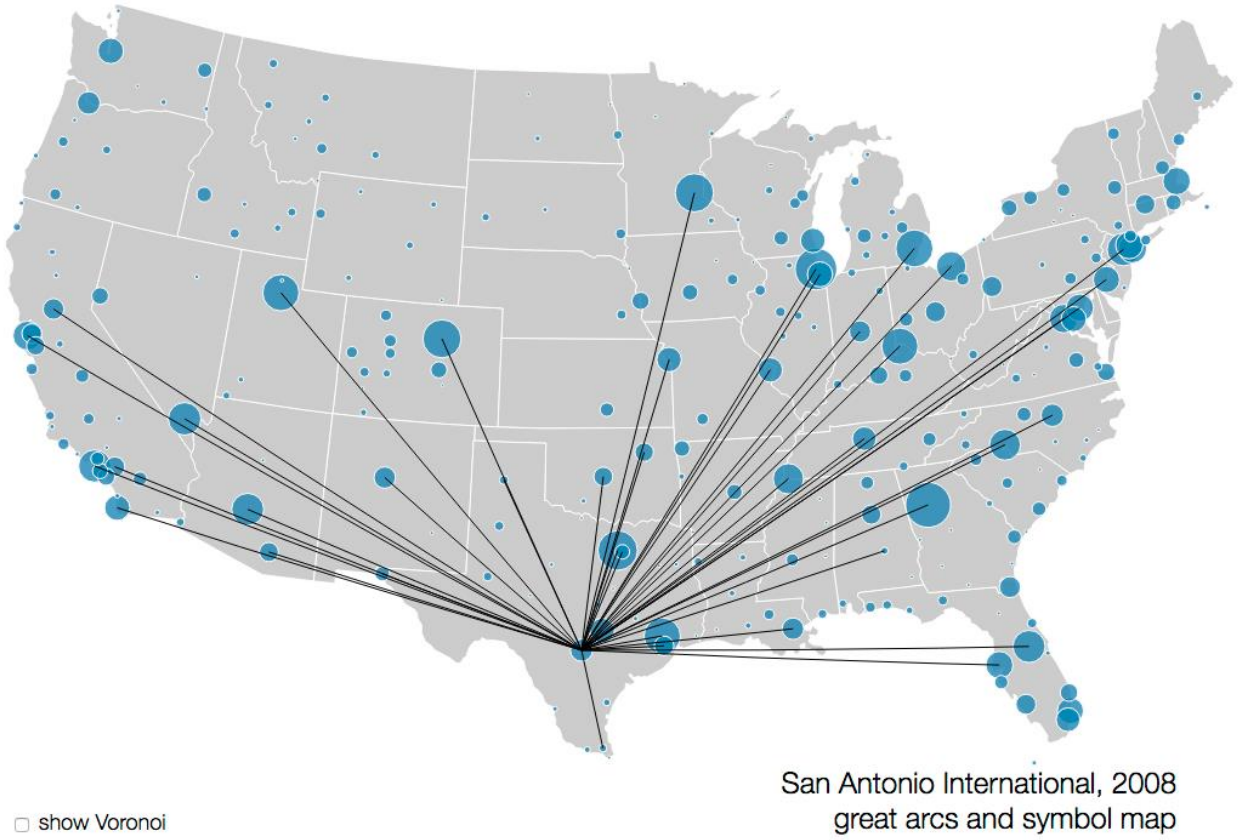


Dynamic Visual Layering

- Interactive based on selection
- One-hop neighbor highlighting demos: click vs hover (lightweight)



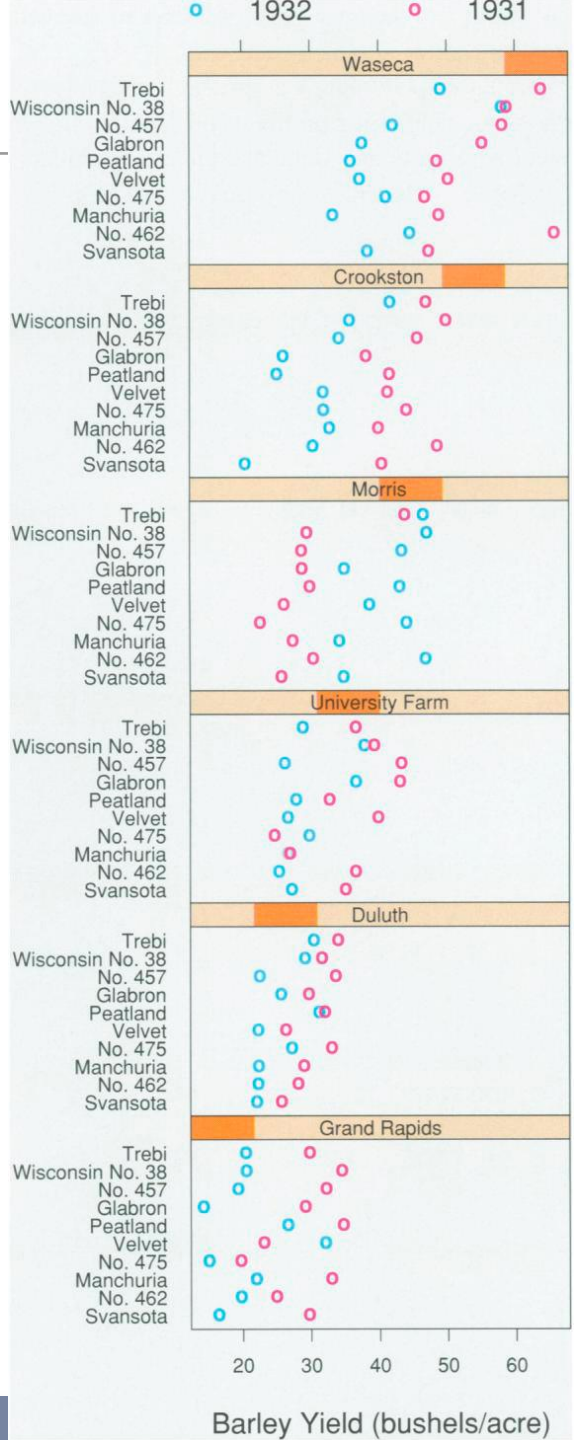
http://mariandoerk.de/edgemaps/demo/



http://mbostock.github.io/d3/talk/20111116/airports.html

Trellis Plots

- Multivariate data is graphed as MxN array of views
  - The same 2 “primary” attributes shown in each view
  - Up to two further variables with MxN discrete values
- Example 1x6 views
  - Different wheat varieties are planted at different sites and the yield (x-axis) per site (N=6 sites), year and wheat type (y-axis) is monitored
  - **Partitioning** → Split by site
  - **Superimpose within same frame** → Color code by year
  - Main-effects ordering
    - Derive value of median for group, use to order
      - Order rows within view by barley variety yield median
      - Order views themselves by site yield median



Quiz

Drag the words into the correct boxes

_____ means placing multiple views side by side. Interaction in such views is usually coordinated, e.g. through _____.

_____ is common for maps to display multiple layers.

Trellis Plots make use of _____ to split the visualization into multiple views.

Juxtapose | Superimpose | Brushing and Linking | Partition

Manipulate

Change

Select

Navigate

Facet

Juxtapose

Partition

Superimpose

Reduce

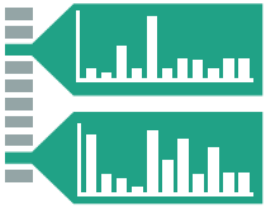
Filter



Aggregate



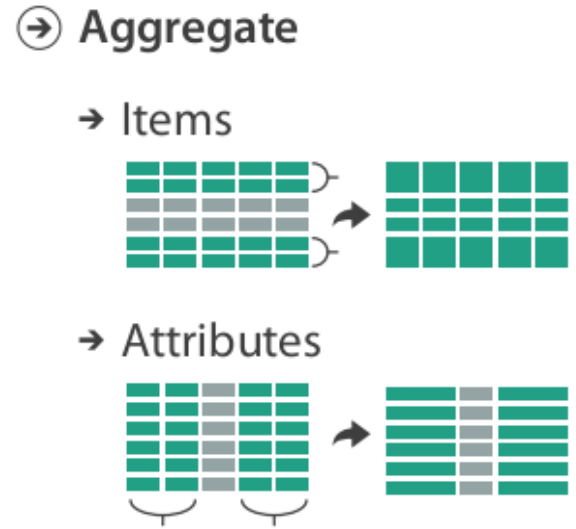
Embed

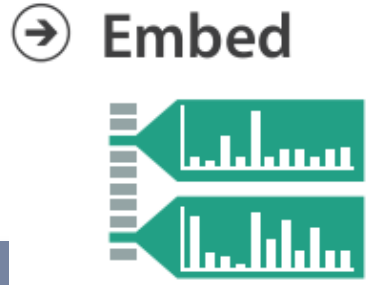


Reduce Items and Attributes

- Filter (data items, attributes)
  - Pro: straightforward and intuitive
    - to understand and compute
  - Con: out of sight, out of mind
- Aggregation
  - Pro: inform about whole set
  - Con: loses information
- Filter and aggregation are not mutually exclusive
  - Combination of filter & show aggregate of rest
  - Combination of facet, reduce, change
- Embed
  - Show detail in context

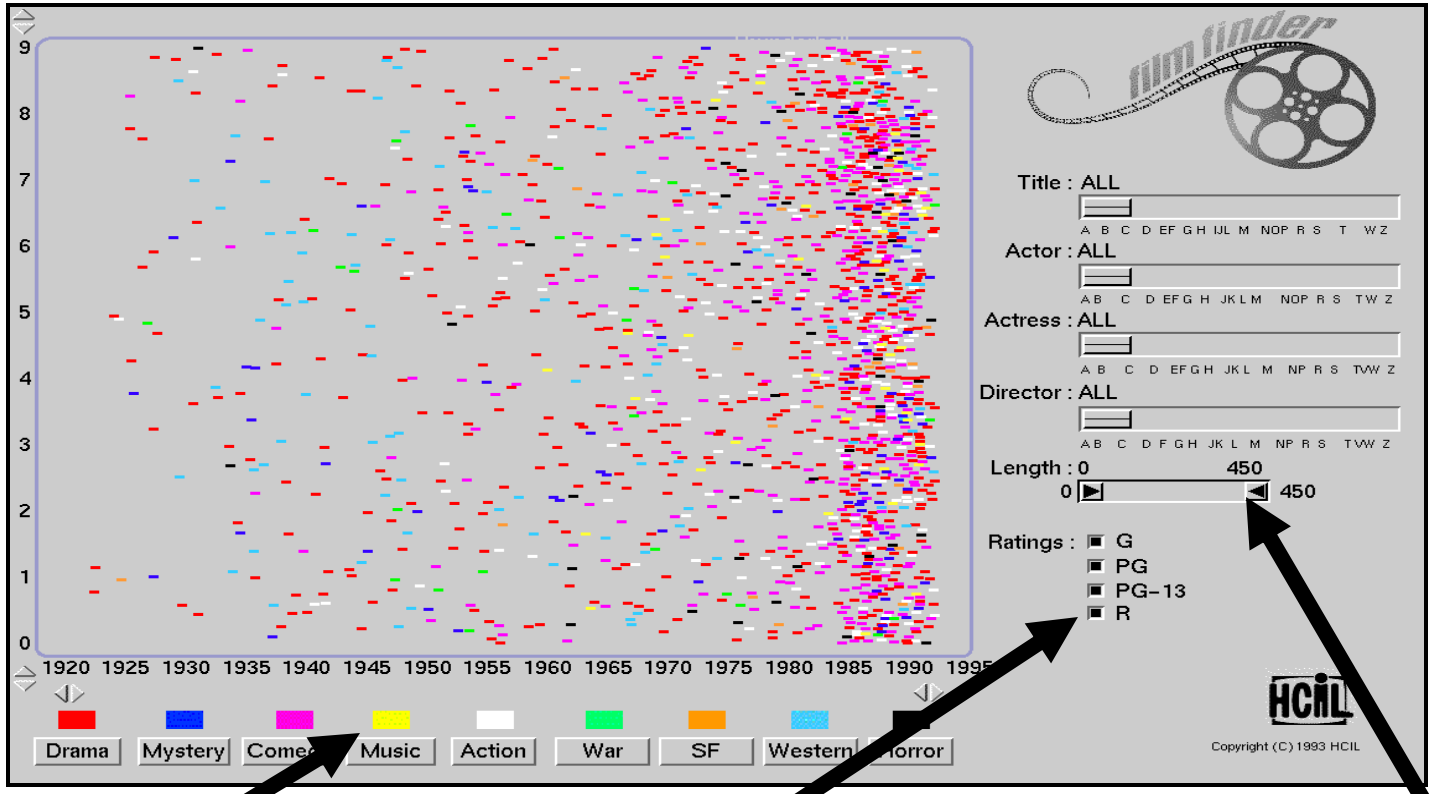






Example System Filmfinder

- Film Finder, University of Maryland



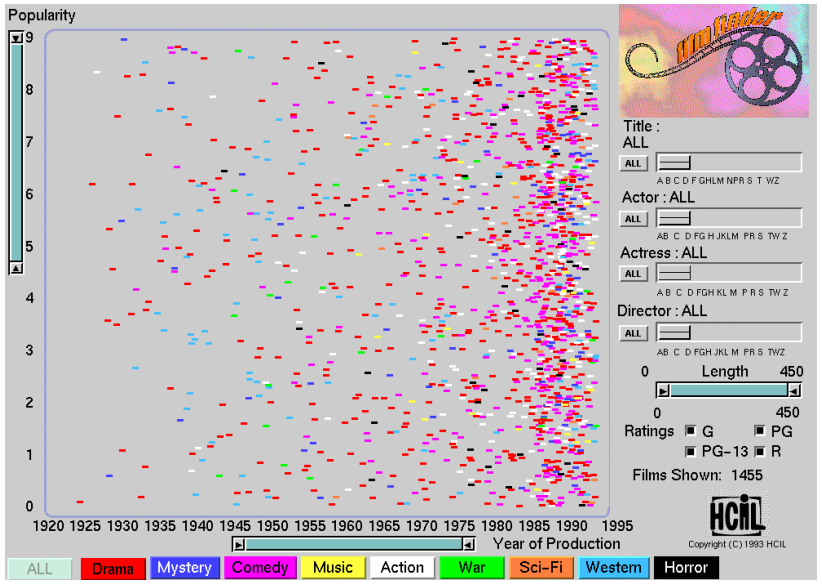
Categorical: Color

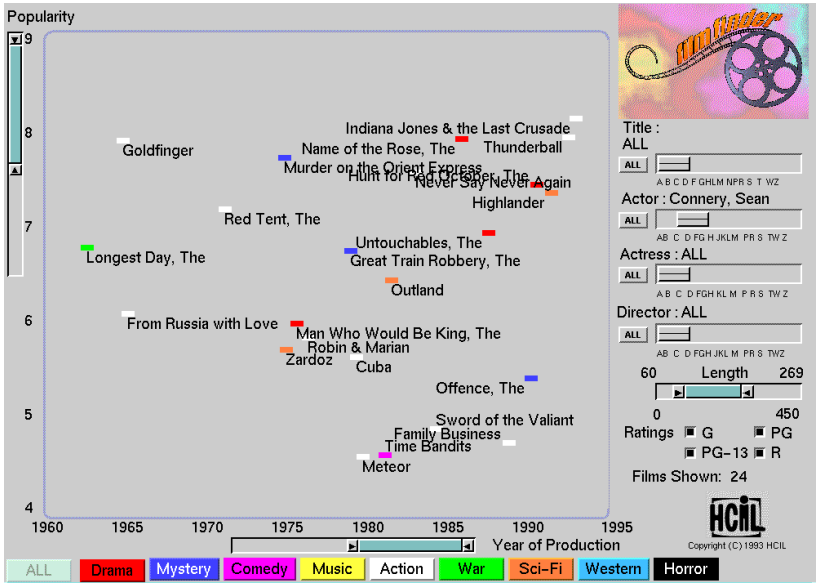
Ordinal: List

Quantitative: Axis

System Filmfinder Supports Dynamic Filtering

- Item filtering
- Browse through tightly coupled interaction
  - Alternative to queries that might return far too many or too few

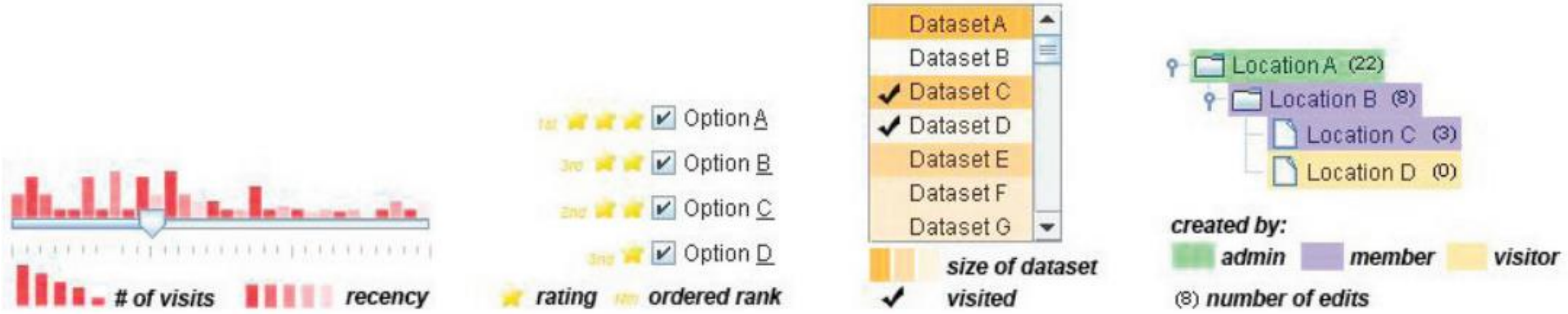




[Visual information seeking: Tight coupling of dynamic query filters with starfield displays. Ahlberg and Shneiderman. Proc. ACM Conf. on Human Factors in Computing Systems (CHI), pp. 313–317, 1994.]

Aggregate Information in Scented Widgets

- The scented widget idiom adds visual encoding information directly to standard graphical widgets to make filtering possible with high information density displays [Willett et al. 07].
- Idea of information scent: cues that help a searcher decide whether there is value in drilling down further into a particular information source, versus looking elsewhere [Pirolli 07].



Embed: Focus + Context

- Combine information within single view
  - Zooming hides the context
  - Two separate displays split attention (e.g. overview+detail
- Elide
  - Drop from view or replace by aggregate information
  - Selectively filter and aggregate
- Superimpose layer
  - Local lens
  - Different semantic layers
- Distortion design choices
  - Region shape: radial, rectilinear, complex
  - How many focus areas: one, many
  - Region extent: local, global

Embed

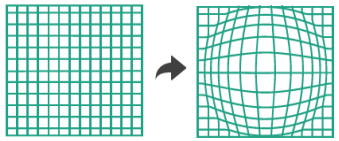
→ Elide Data



→ Superimpose Layer



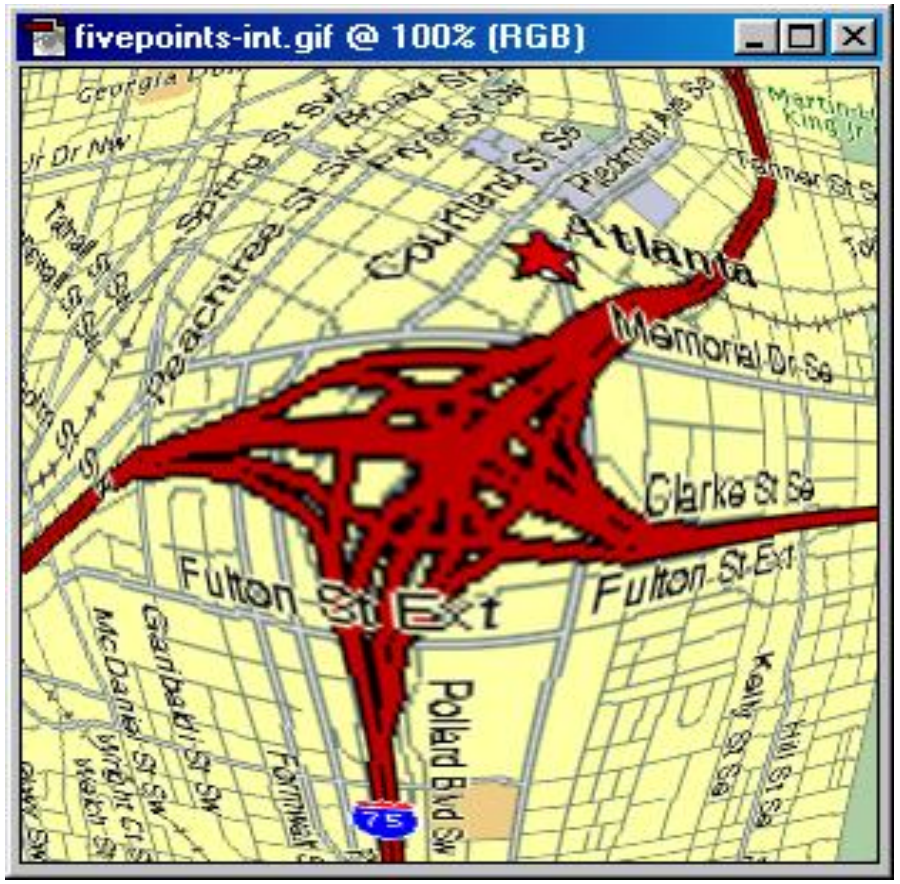
→ Distort Geometry



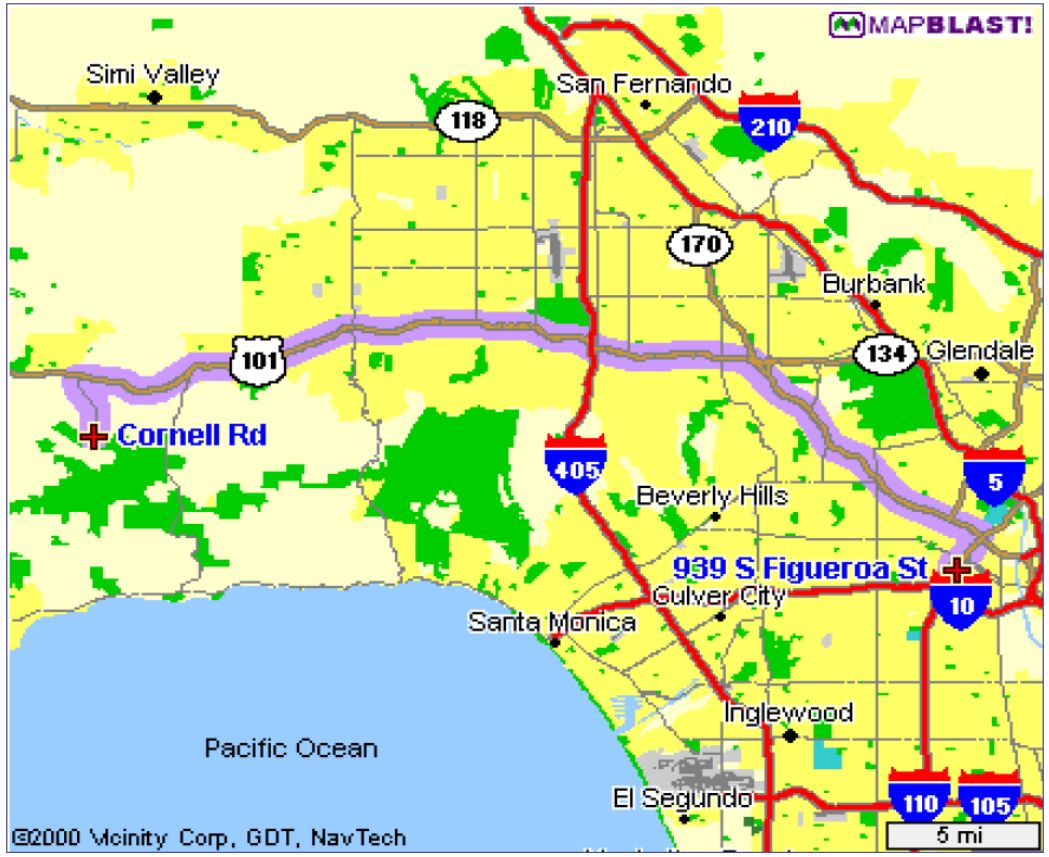
Distort Geometry: Fish-eye View

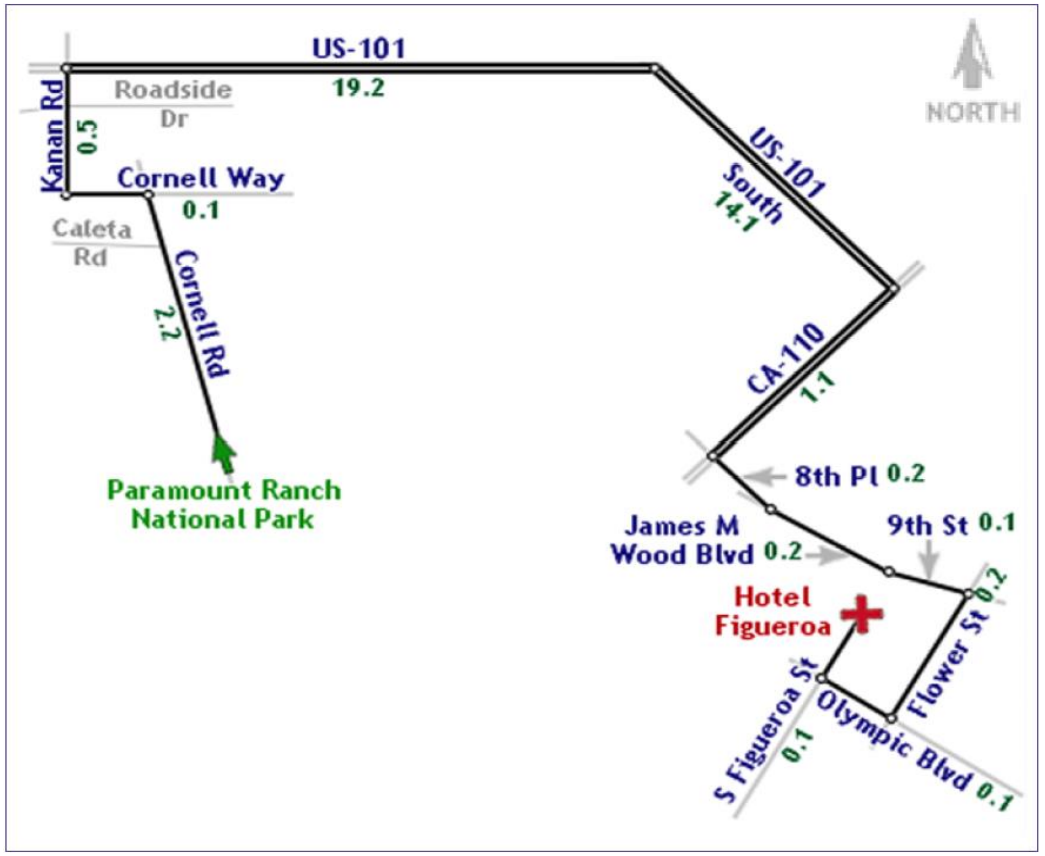
- Magnified image is distorted so that focus is at high magnification, periphery at low
- All in one view
- Distortion can be disorienting

- This is focus + context
  It is **not** called
  OVERVIEW and DETAIL



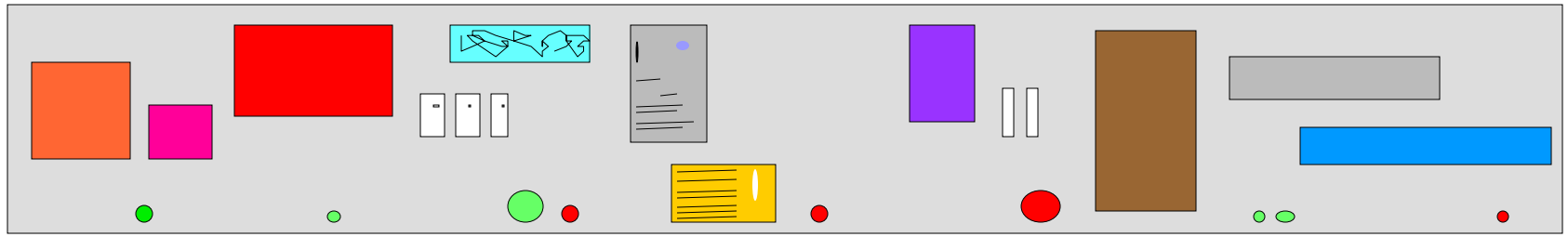
Effective Route Maps: Two Focus Regions



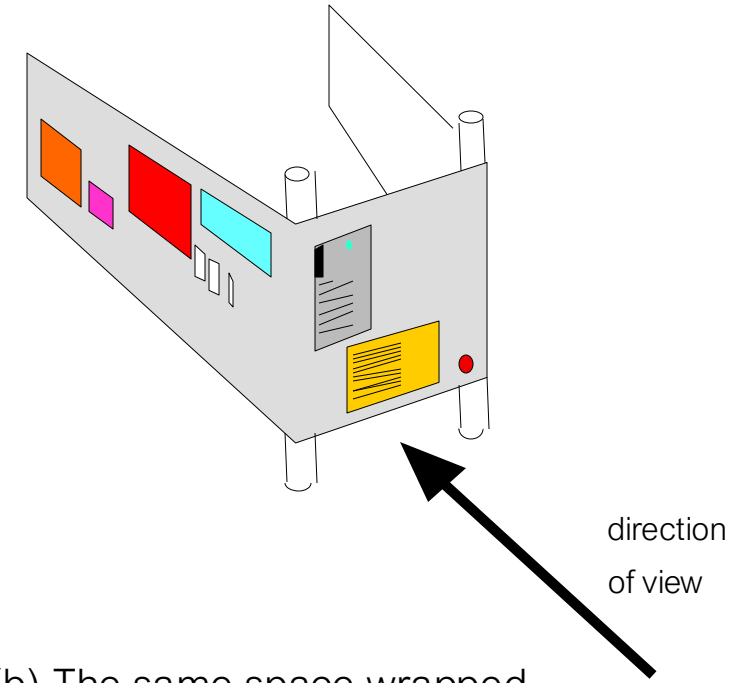


Agrawala/Stolte, Siggraph 2001:
Rendering Effective Route Maps: Improving Usability Through Generalization

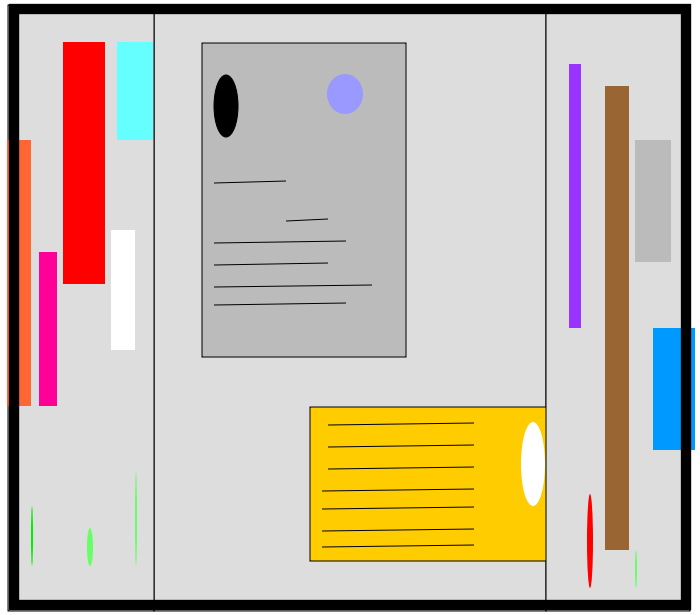
Bifocal Display (Spence & Apperly ‘82)



(a) An information space containing documents, emails, etc.

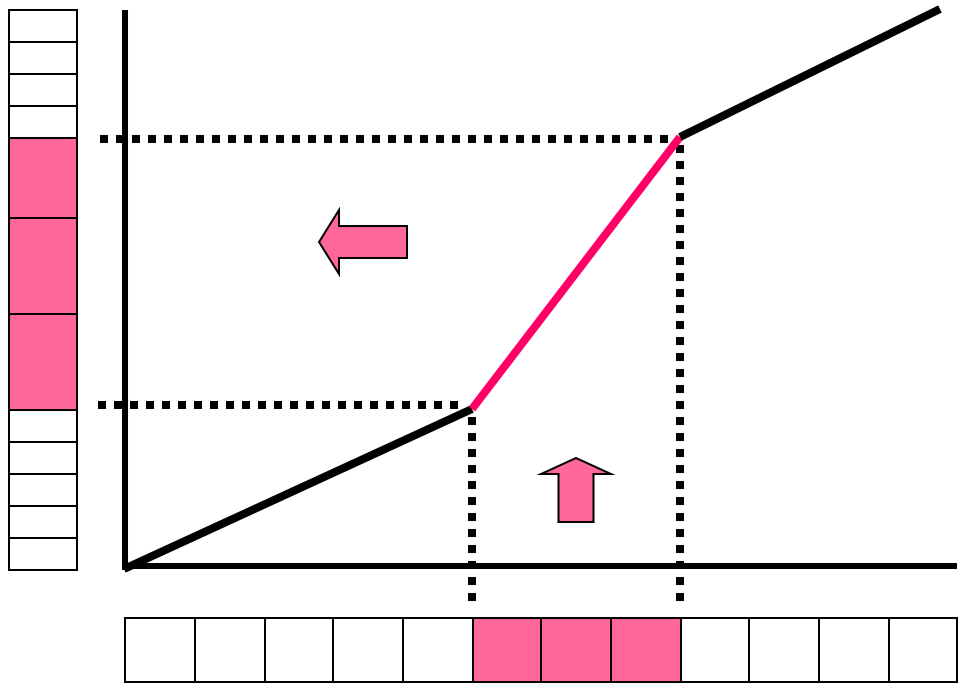


(b) The same space wrapped around two uprights.



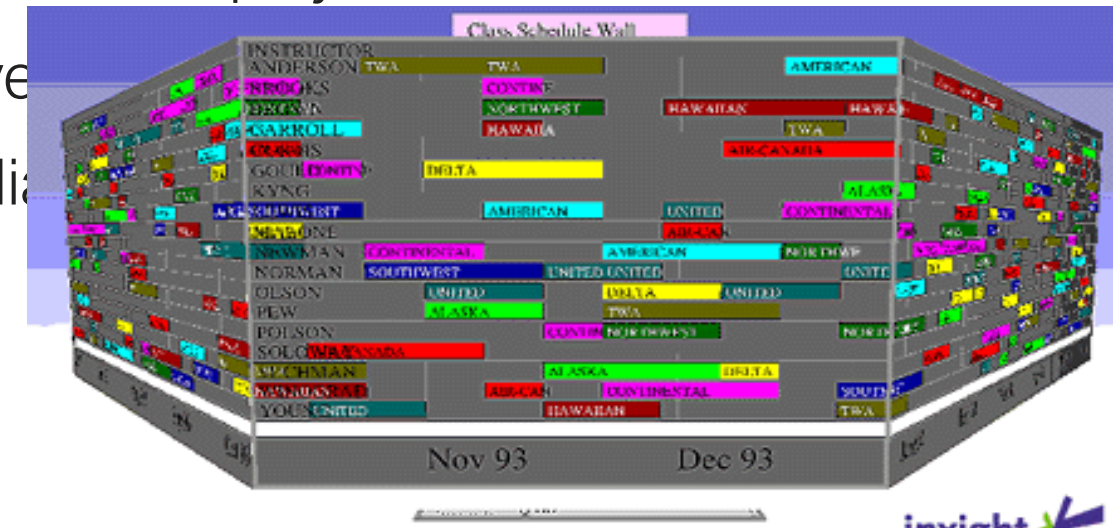
(c) Appearance of the information space when viewed from an appropriate direction

Visual Transfer Functions



Perspective Wall

- 3D implementation of bifocal display
- Uses natural perspective
- Map work charts onto di
  - x-axis is time
  - y-axis is project



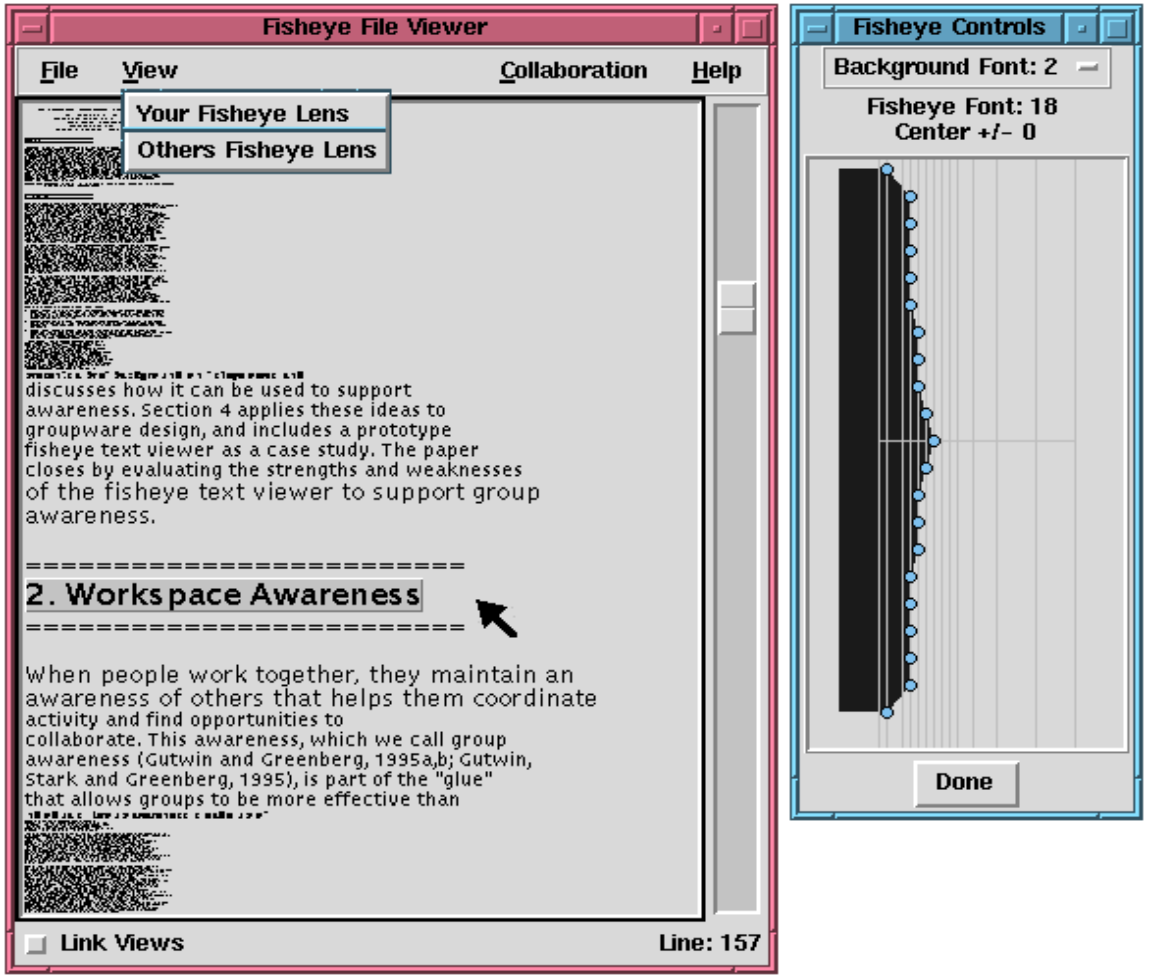
inxight

Mackinlay, Robertson, Card CHI'91

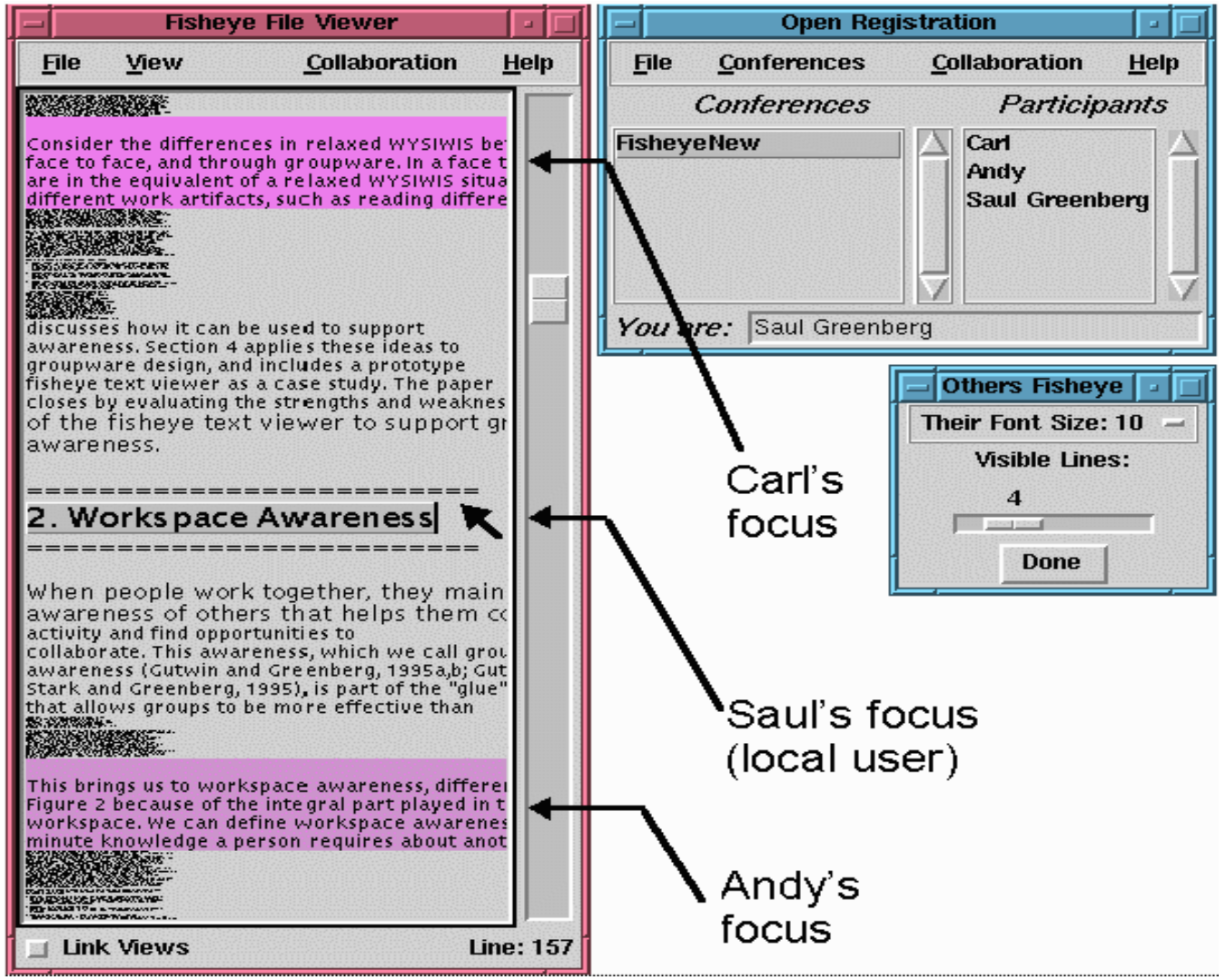
Focus and Context: Text/Program Viewing

Shown here are examples from Gutwin and Greenberg

Step function

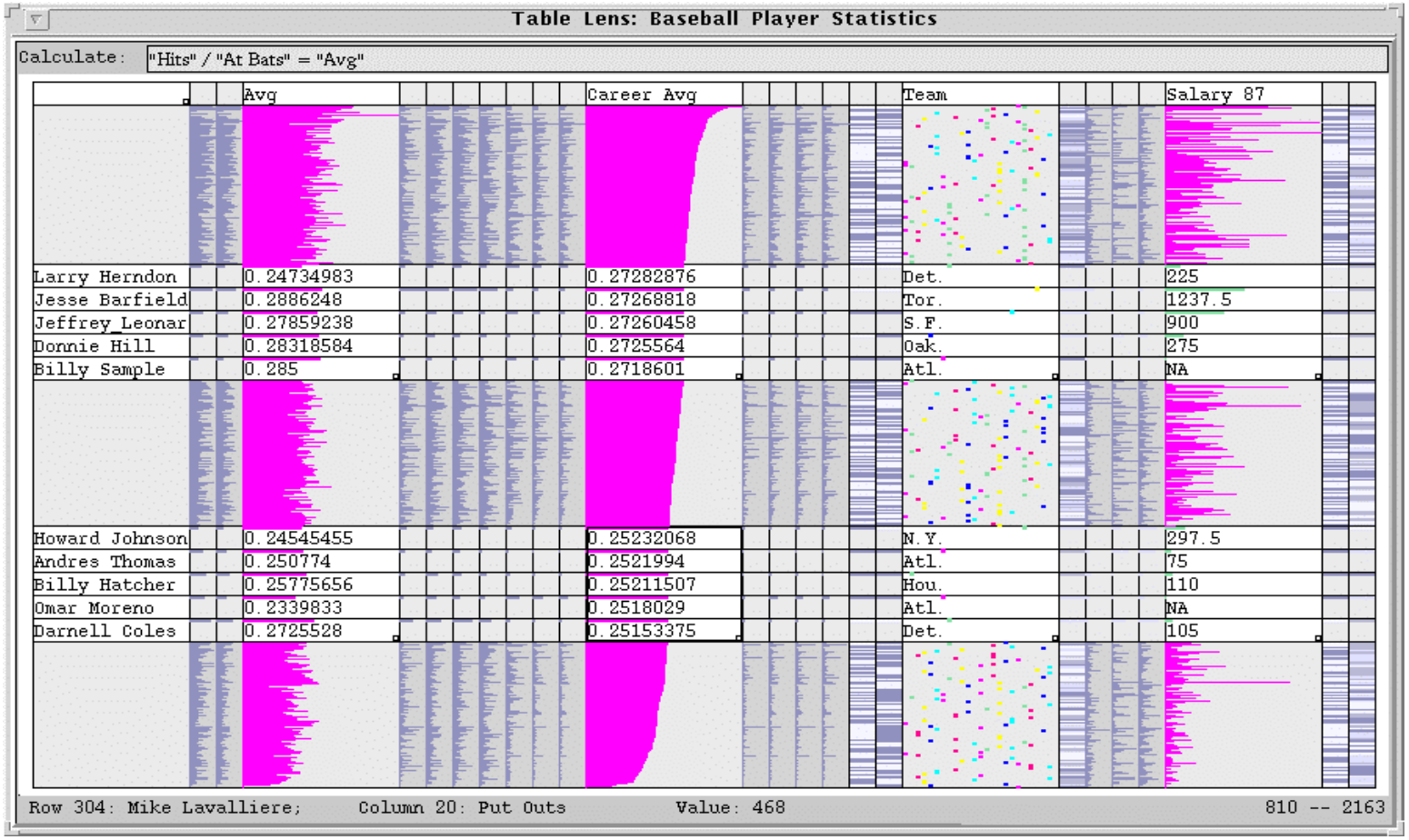


Focus and Context: Shared Text editor for CSCW



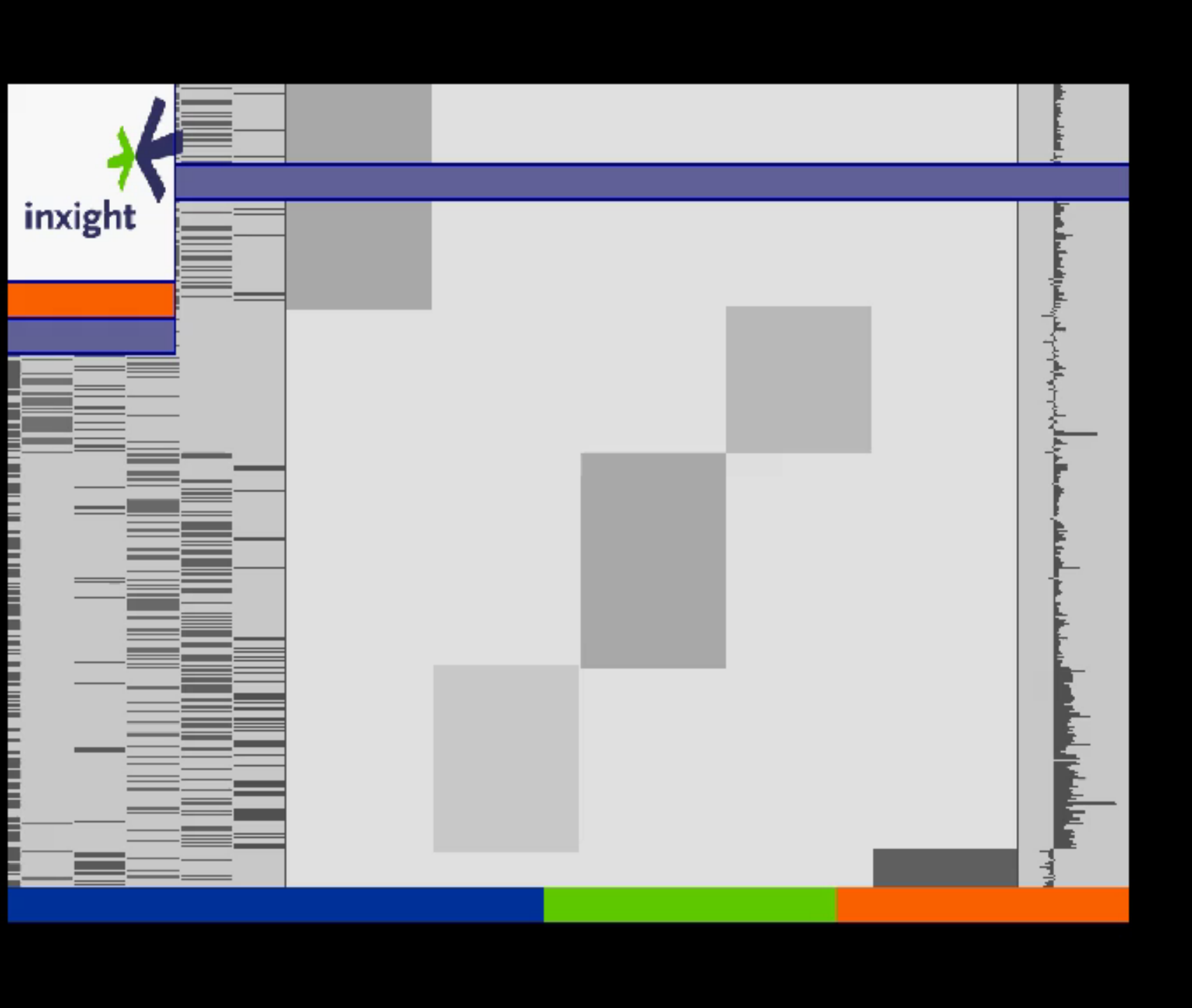
Gutwin and Greenberg, '96

Focus and Context Example: Eureka



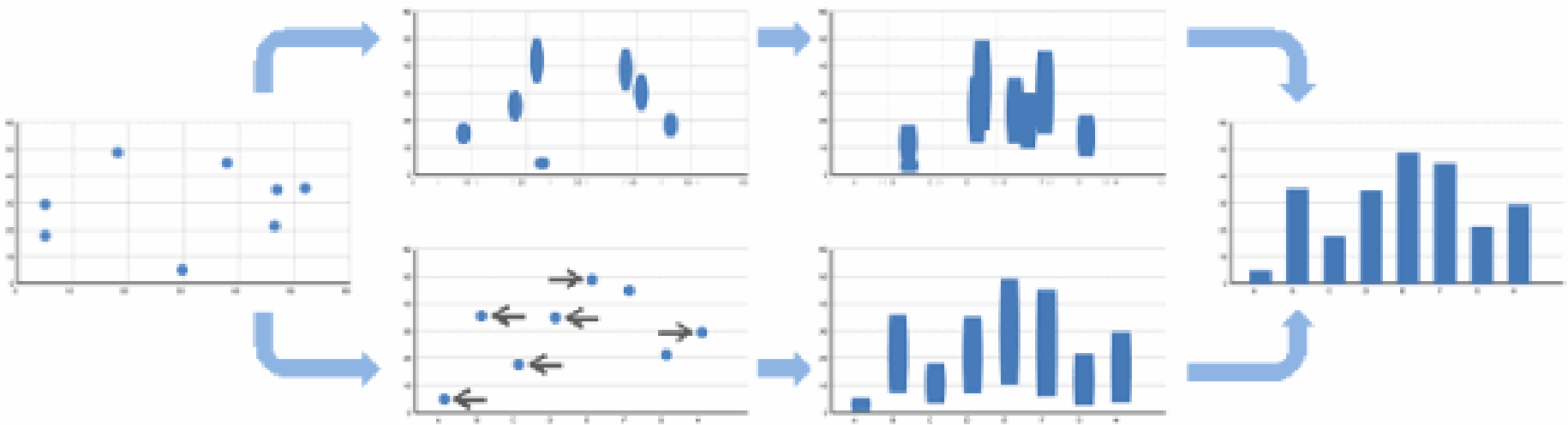
From Xerox PARC and Inxight

http://www.inxight.com/products/sdks/tl/


inxight

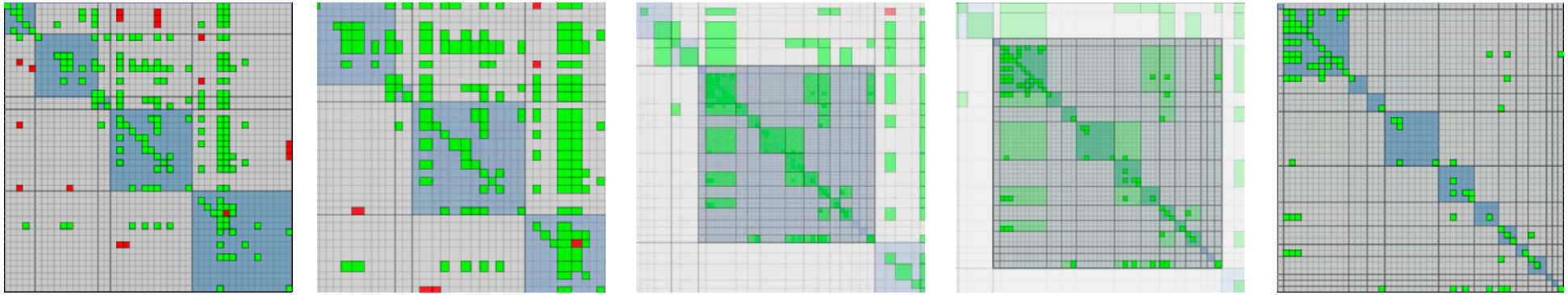
Important for Focus and Context: Animated Transitions

- Smooth transition from one state to another
  - Alternative to jump cuts
  - Support for item tracking when amount of change is limited
  - Focus should open, shift and close in an animated way
  - http://vis.berkeley.edu/papers/animated_transitions/



Important for Focus and Context: Animated Transitions

- Animated transitions are useful in general
  - Example: multilevel matrix views
    - Scope of what is shown narrows down
      - Middle block stretches to fill space, additional structure and detail appears within
      - Other blocks squish down to increasingly aggregated representations



[Using Multilevel Call Matrices in Large Software Projects. van Ham. Proc. IEEE Symp. Information Visualization (InfoVis), pp. 227–232, 2003.]

Quiz

Tick only correct statements.

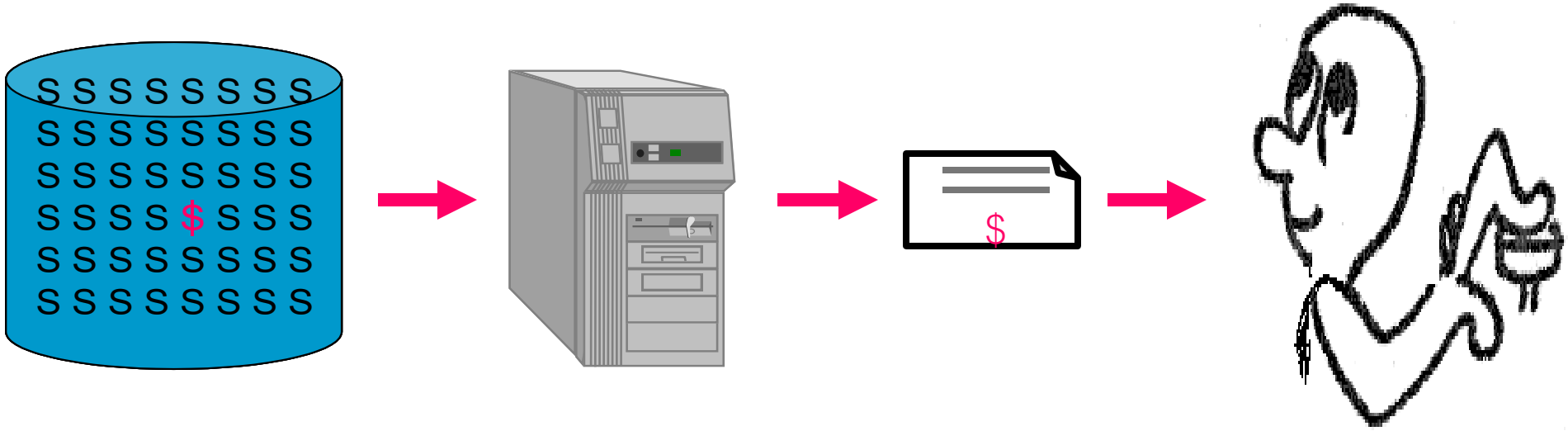
- [ ] Animated transitions help viewers to keep track of items.
- [ ] The fish-eye view is a form of *Overview and Detail*.
- [ ] Visualizations can only implement filter or aggregation techniques but not both.
- [ ] Aggregation leads to a loss of information.

End

Data Space

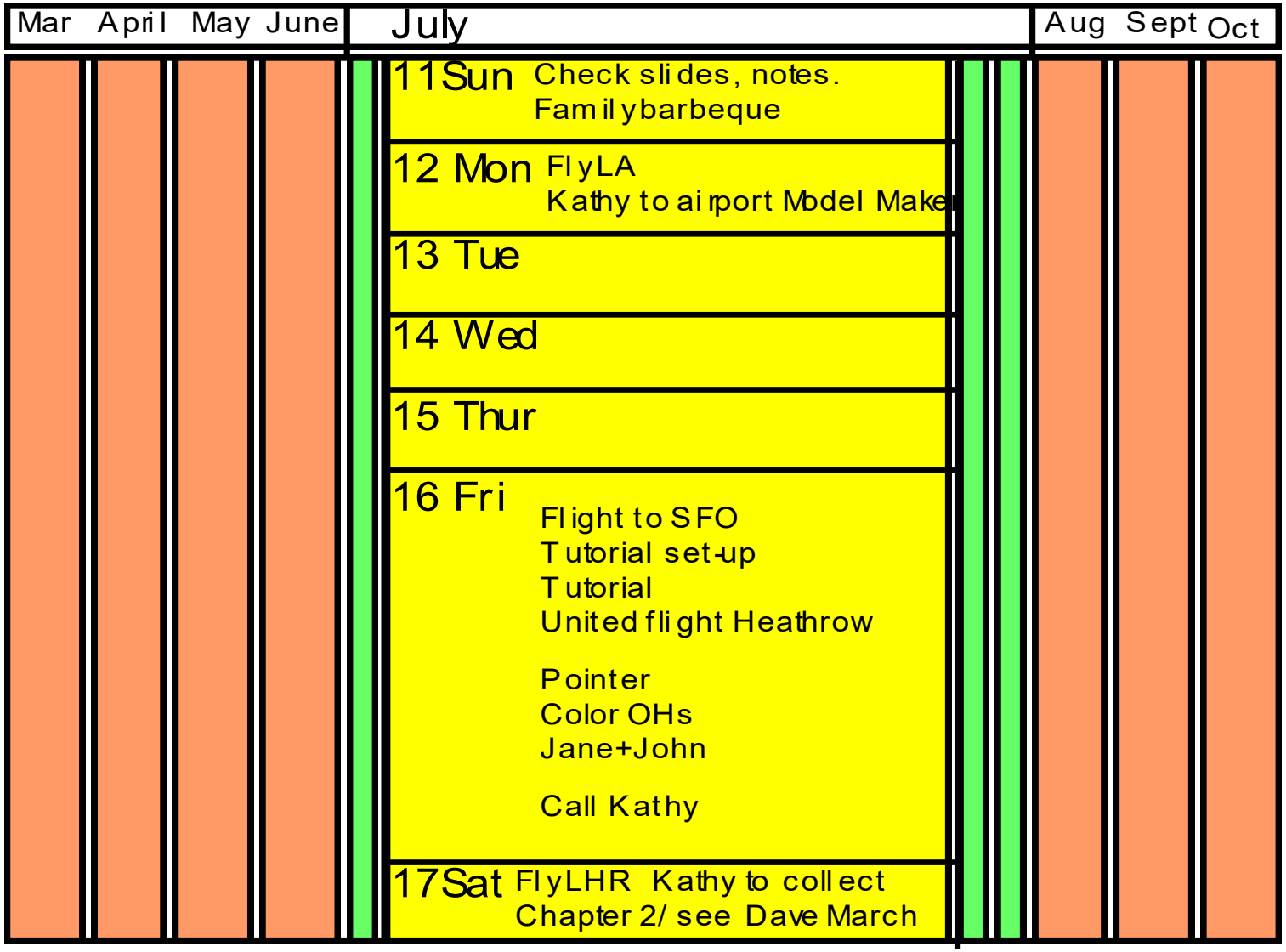
- Statistical distributions: multi-dimensional databases,…
- Trees/Hierarchies: directories,…
- Networks/Graphs: web, communications,…
- Unstructured data, e.g. document collections: digital libraries,…
- Time Series
- Maps

Competing Approach: Data Mining

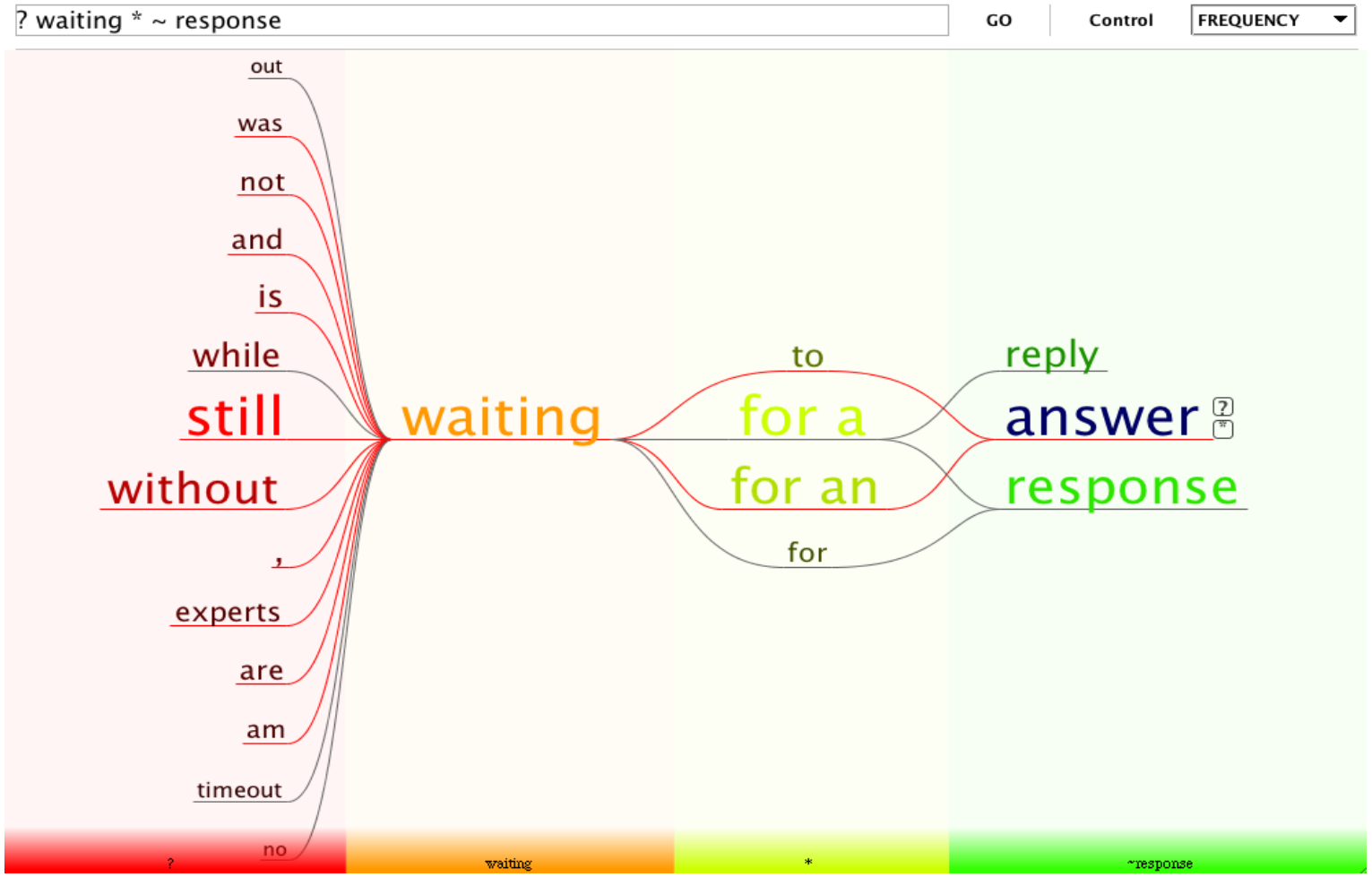


- Tap the power of the computer
  - Complex statistical analysis
  - Simple report
  - Visual Analytics: combine data mining and info vis

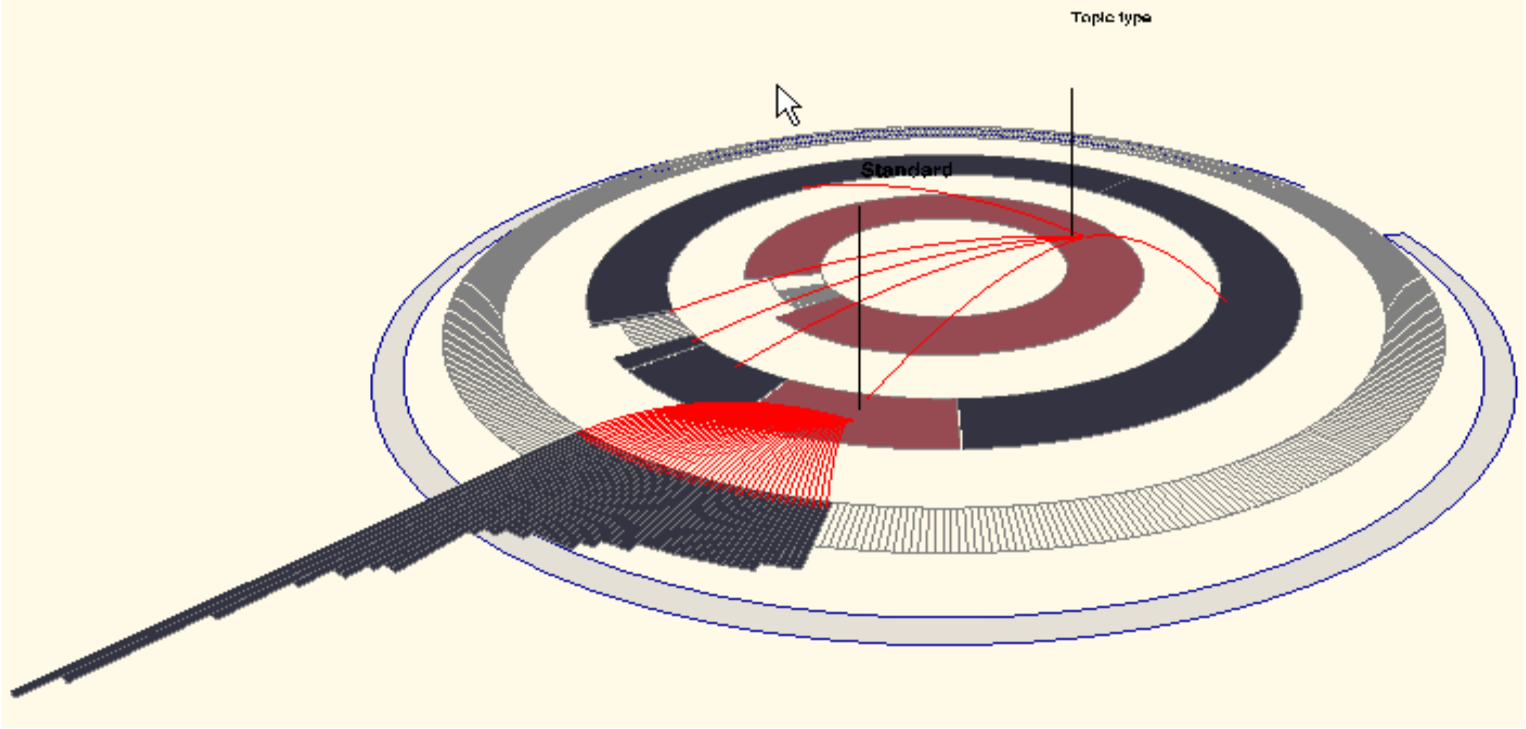
A First Bifocal Calendar (1980)



Visual Analytics: Netspeak Wordgraph



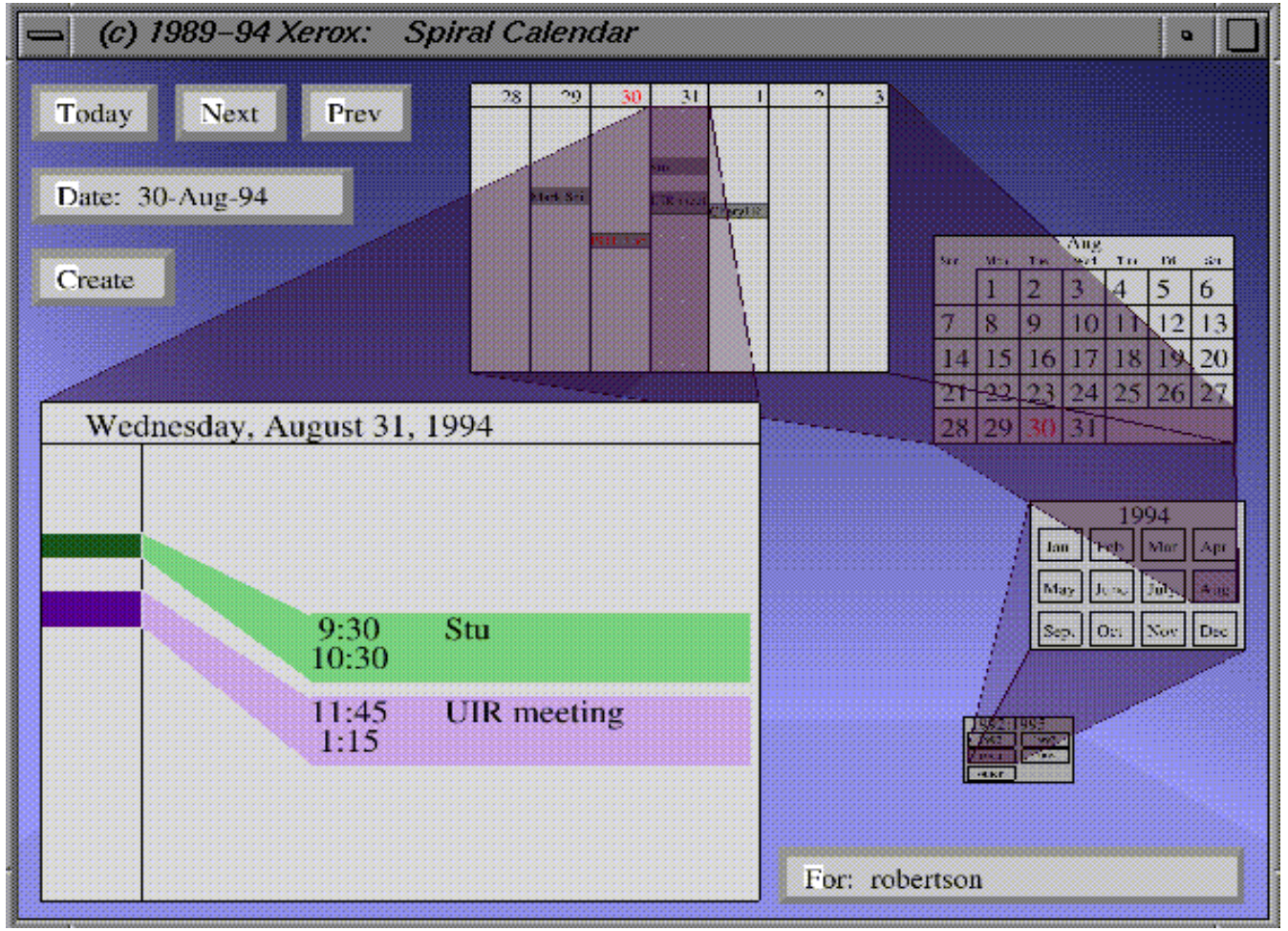
The i-disc



Perspective: Natural focus and context display

Spiral Calendar

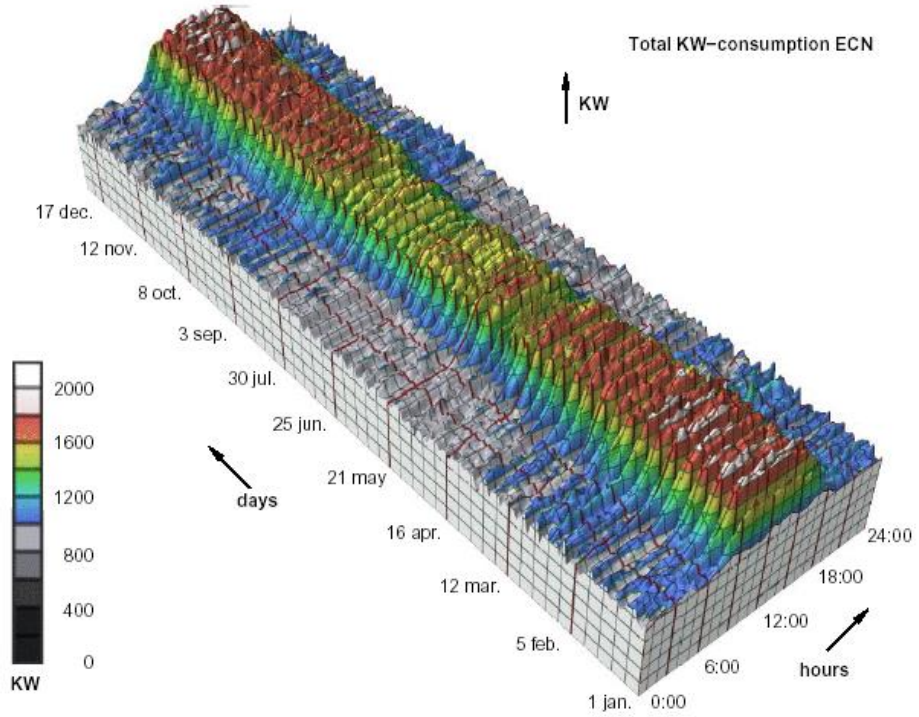
This is more Detail and Overview More Than Focus and Context



Mackinlay, Robertson, Deline UIST'94

No unjustified 3D example: Time-series data

- extruded curves: detailed comparisons impossible

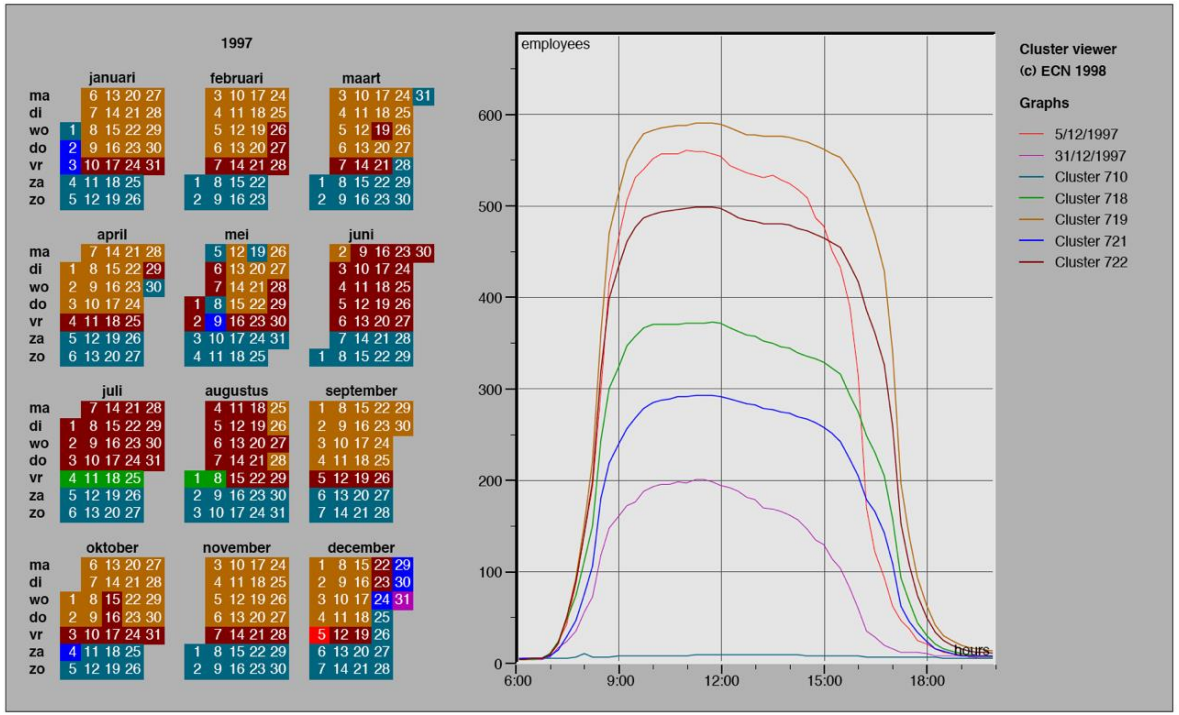


[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.]

64

No unjustified 3D example: Transform for new data abstraction

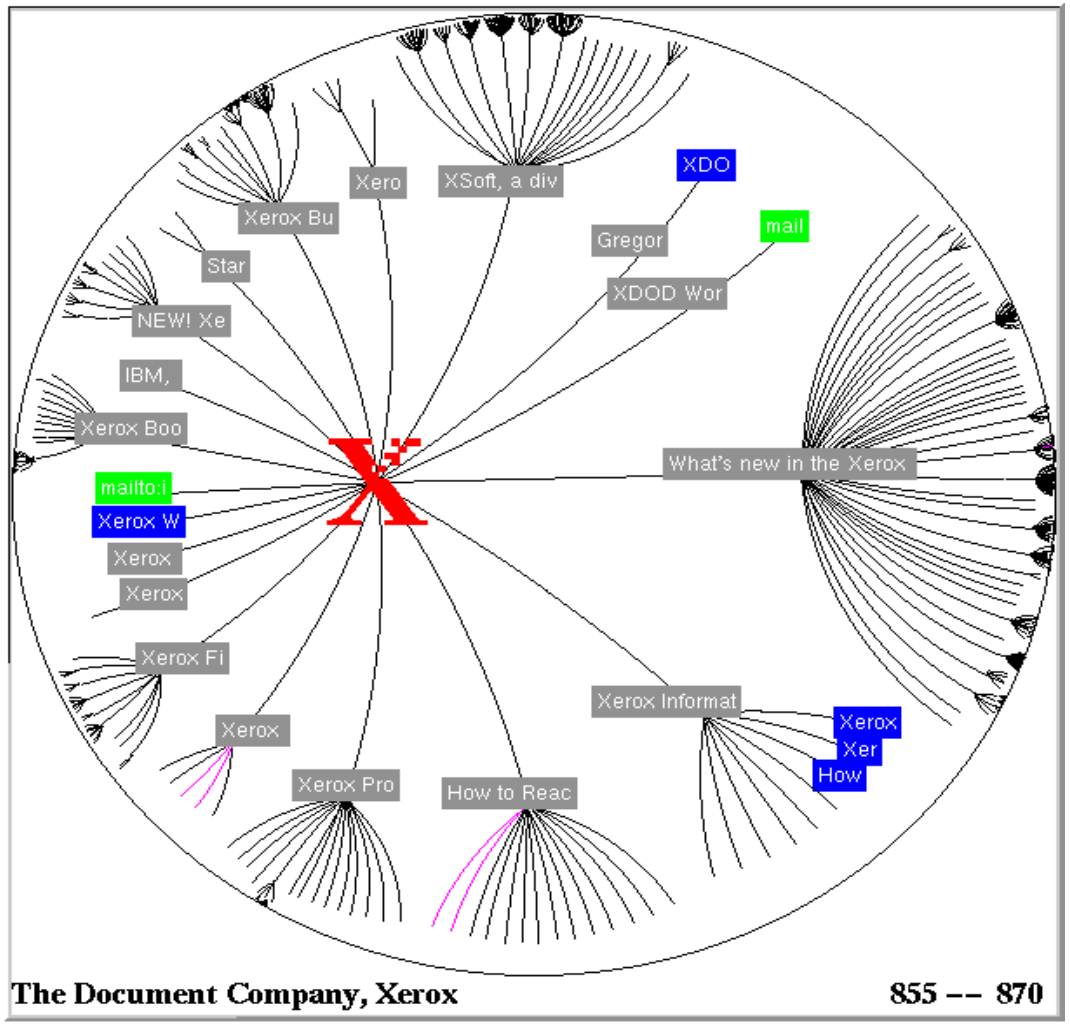
- derived data: cluster hierarchy
- juxtapose multiple views: calendar, superimposed 2D curves



[Cluster and Calendar based Visualization of Time Series Data. van Wijk and van Selow, Proc. InfoVis 99.]

65

Hyperbolic Browser



DEMO

Commercial version

Star Tree

http://www.inxight.com

Now: SAP!

Lamping, Rao UIST'94

Fundamental Problem

- Scale: Most datasets don’t fit on the display
  - Too many attributes
  - Too many data records
- Therefore, user must focus on specific attributes and/or data records
- Issues raised by focusing:
  - How to specify focus?
  - What is the relationship of focus to the rest of the data?
  - How does the user shift focus?
    - Find new focus
    - Stay oriented

Example Filmfinder: Data Types

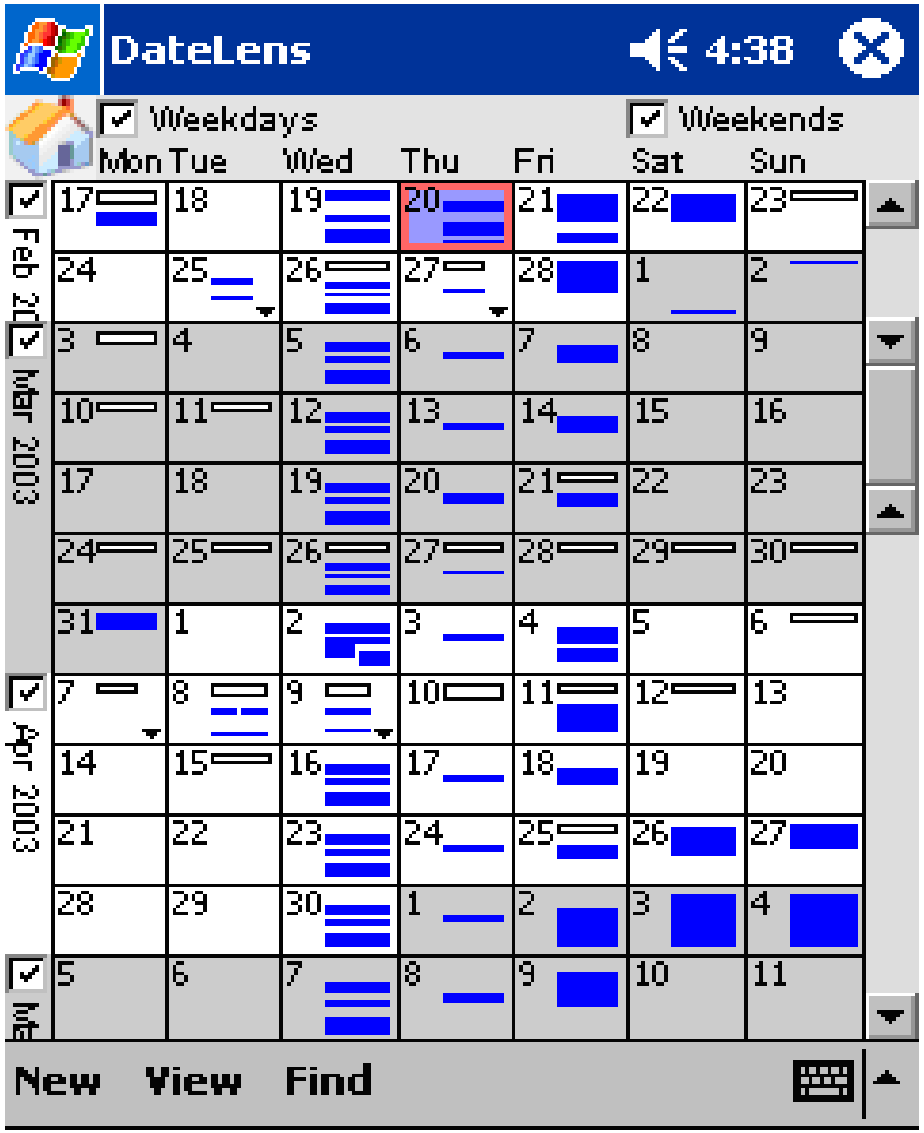
Categorical

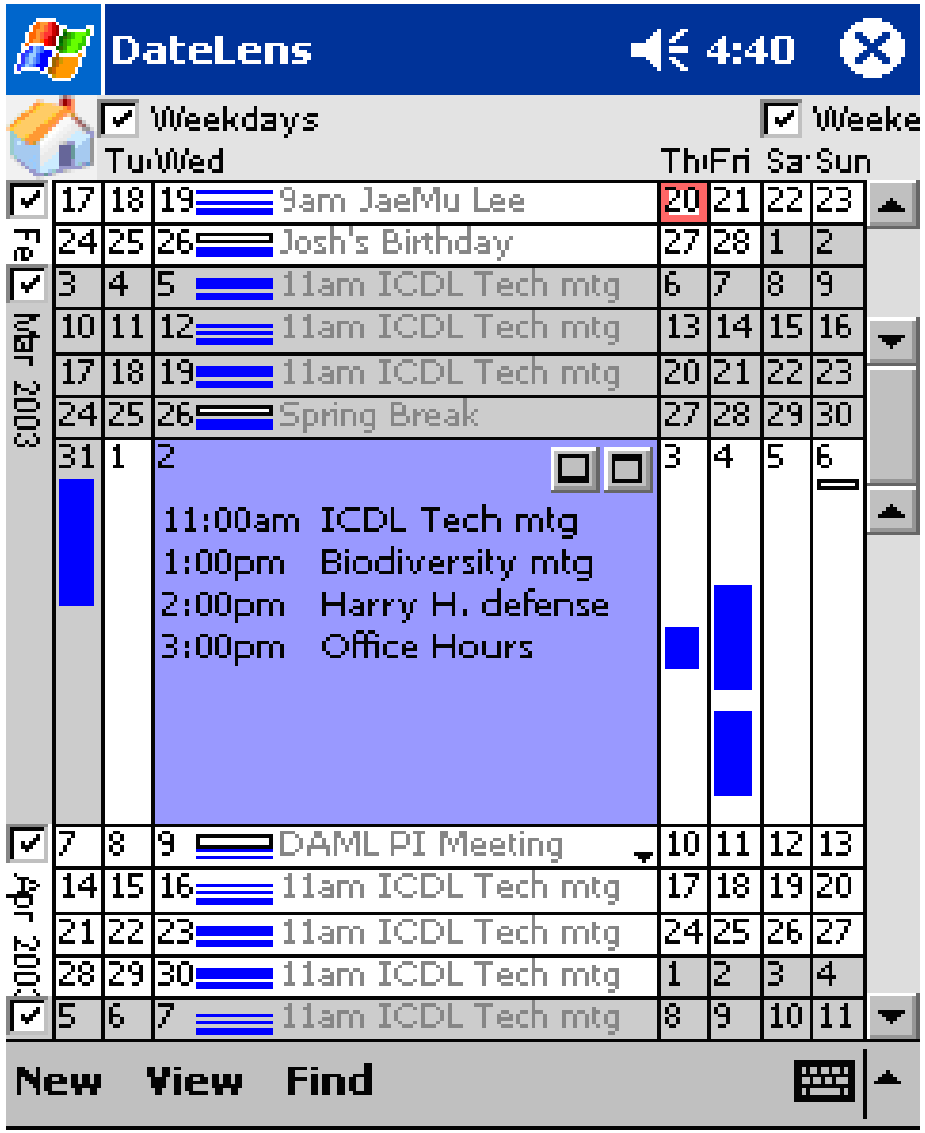
Ordinal

Quantitative

FilmID	230	105	540	…
Title	Goldfinger	Ben Hur	Ben Hur	…
Director	Hamilton	Wyler	Niblo	…
Actor	Connery	Heston	Novarro	…
Actress	Blackman	Harareet	McAvoy	…
Year	1964	1959	1926	…
Length	112	212	133	…
Popularity	7.7	8.2	7.4	…
Rating	PG	G	G	…
Film Type	Action	Action	Drama	…

Bifocal Calendar: Fishcal / DateLens



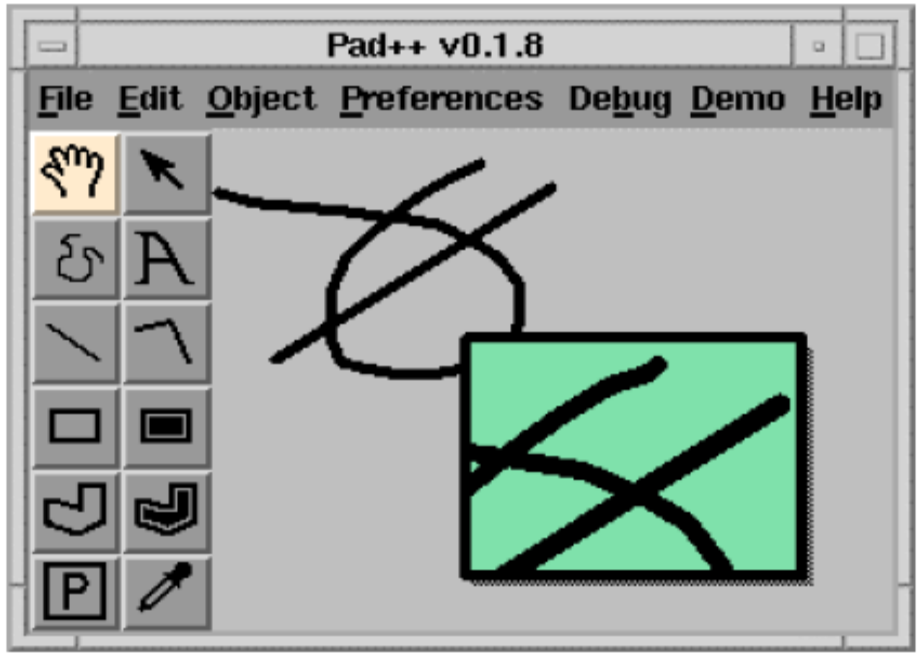


Important Concepts

- Portals
- Lenses
- Sticky objects
- Semantic zooming
- Aggregation and abstraction

Portals

- Views onto another place in the world
- Implemented typically as separate rectangular region
- Zooming, panning, I/O all work independently
- Can be used to create overviews or focus re

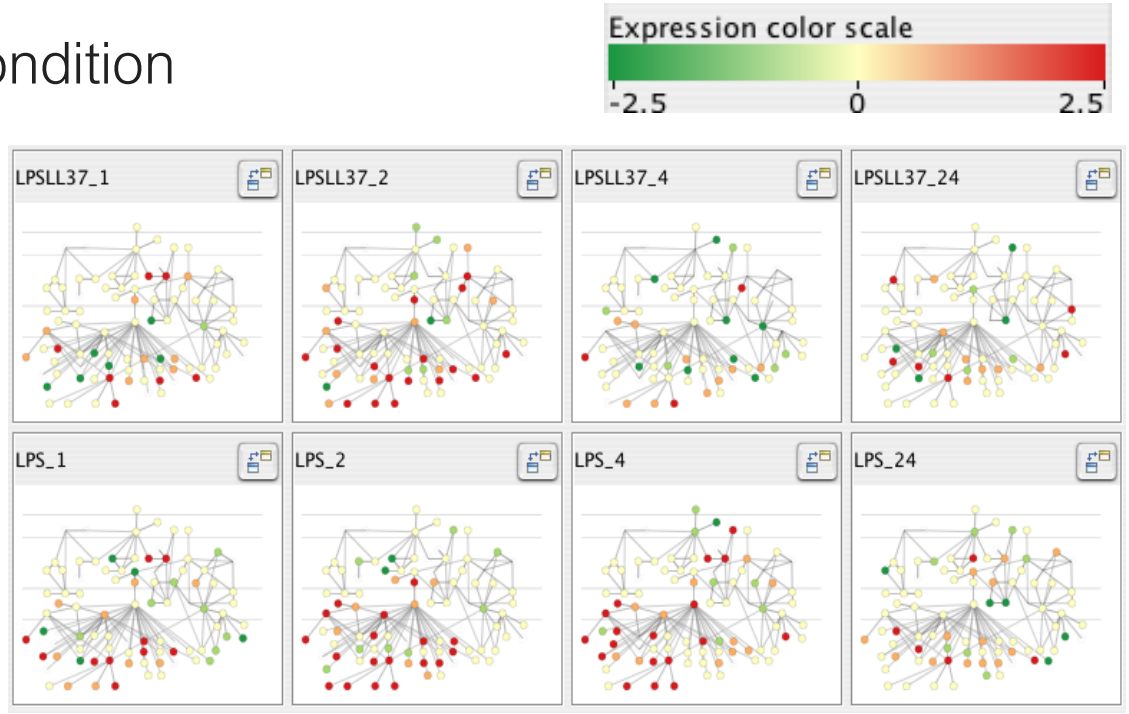


Sticky Objects

- Objects in the world that do not respond to the basic zoom/pan interface physics
  - E.g. coordinate display, counter, …
- Objects are “stuck” to the display
  - They never change position
  - They never change size

Eyes beat memory example: Cerebral

- Small multiples: one graph instance per experimental condition
  - same spatial layout
  - color differently, by condition



[Cerebral: Visualizing Multiple Experimental Conditions on a Graph with Biological Context. Barsky, Munzner, Gardy, and Kincaid. IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis 2008) 14:6 (2008), 1253–1260.]

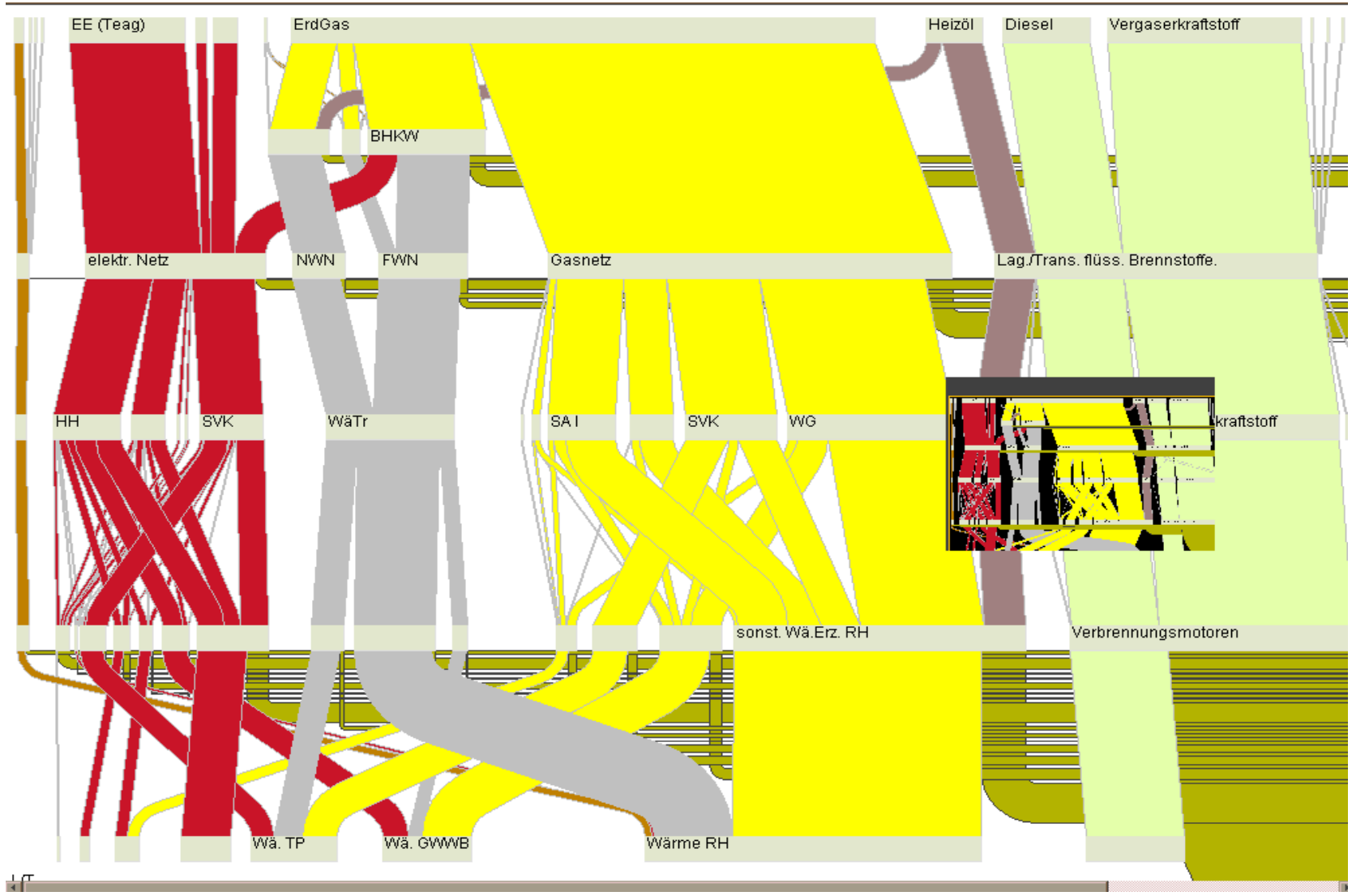
Why not animation?

- disparate frames and regions: comparison difficult
  - vs contiguous frames
  - vs small region
  - vs coherent motion of group

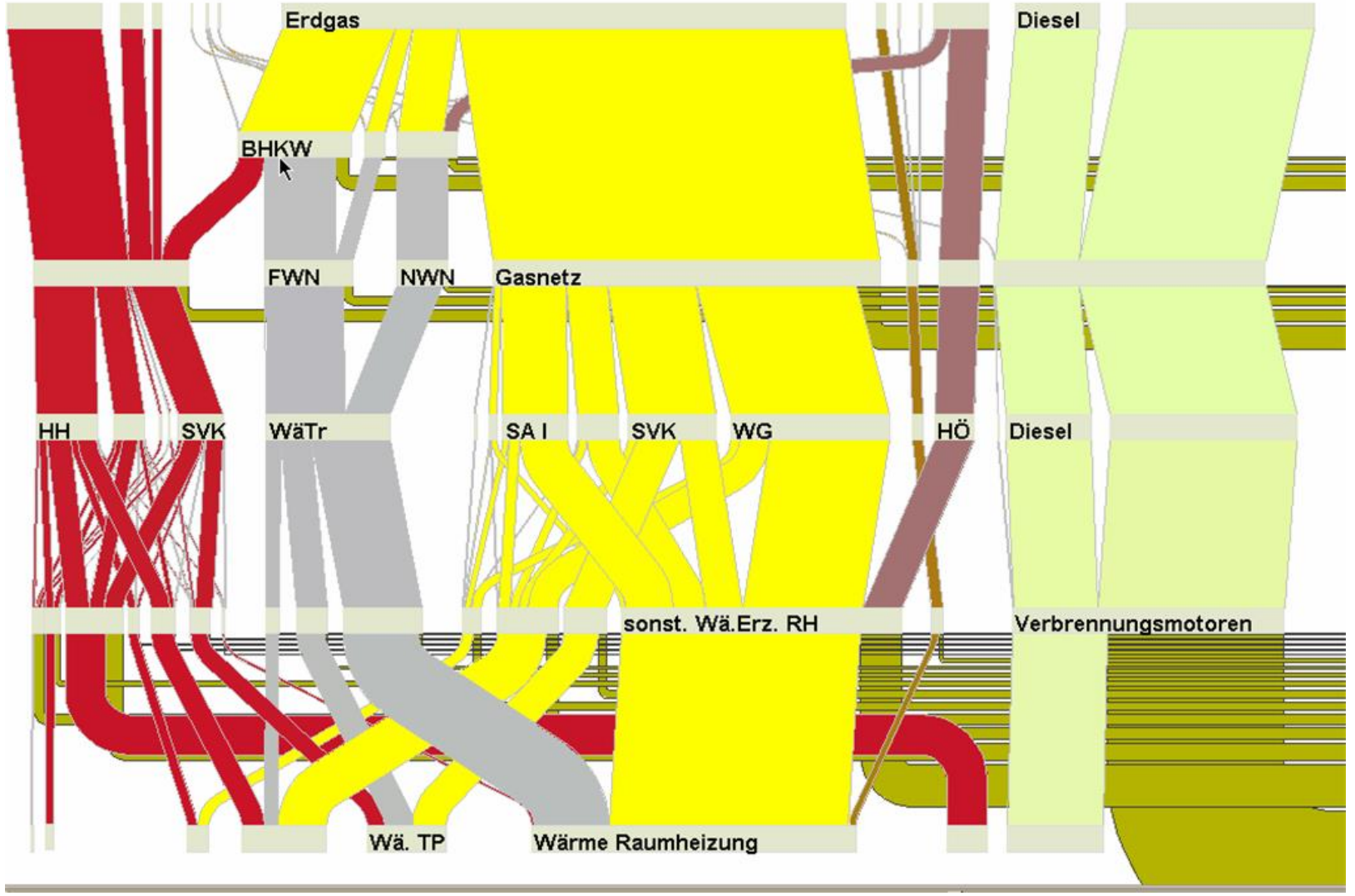
- change blindness
  - even major changes difficult to notice if mental buffer wiped

- safe special case
  - animated transitions

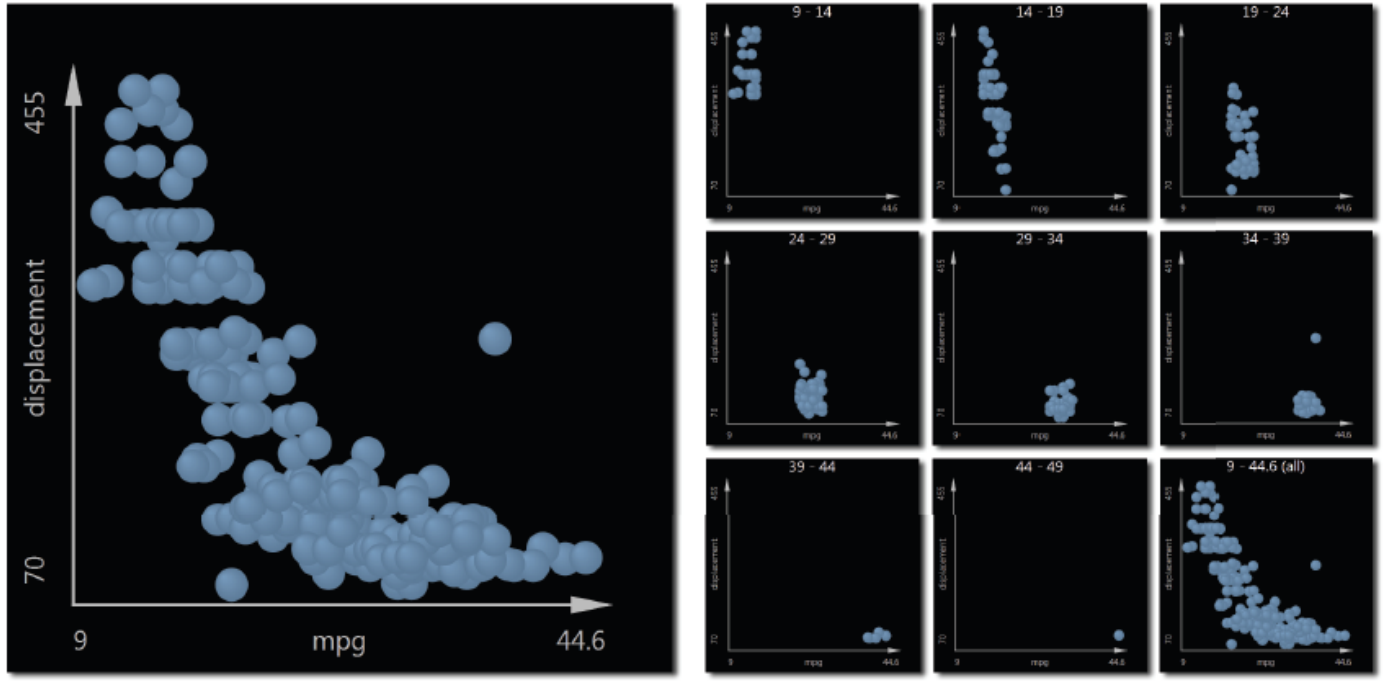
Aggregation Example



Aggregation Example

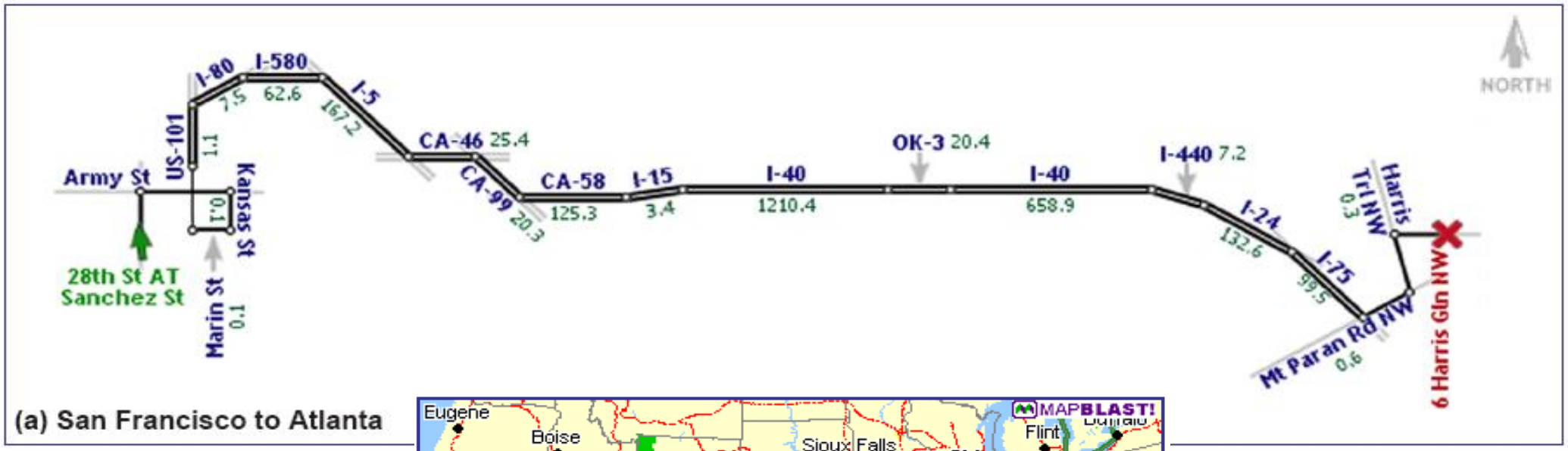


Small Multiples and Large Singles

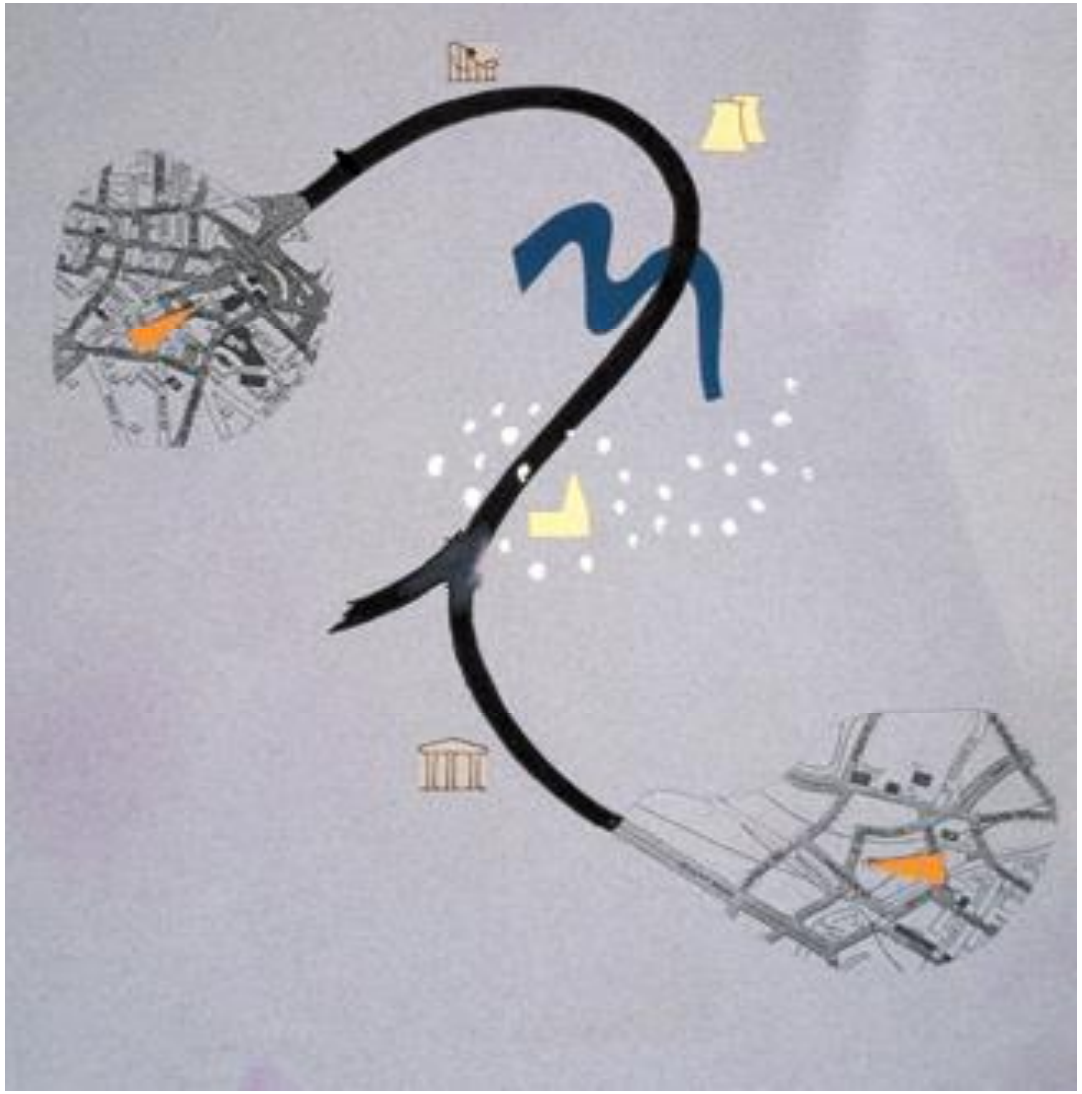


https://vimeo.com/86688905

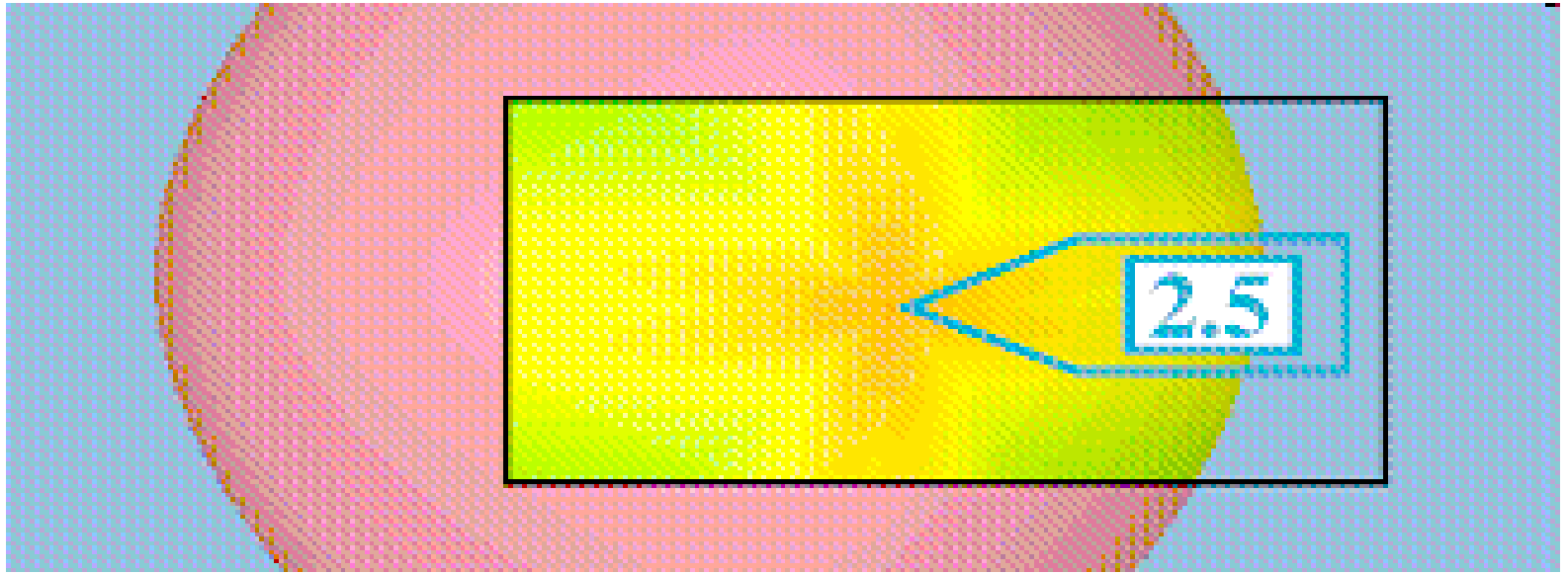
Another Route Map Example



A Really Useful Map: Two Focus Regions



Superimpose Local Lens: Magic Lenses



Gaussian curvature pseudo-color lens
Widget to read the value of the curvature